

Sphere and Schur sources for the localized Weil form

Exact local norms and the inverse bulk matching problem

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Abstract

We study the problem of realizing the complete localized Weil quadratic form as the norm of a linearly prepared state in an independently positive quantum or boundary system. The intended difficulty is an exact matching identity, potentially conditional on construction of a specified field theory. We give explicit calculations in sphere and Schur quantization. The real Gaussian boundary module supplies the quarter-shift gamma factor and the positive oscillator tower, while its natural dilation pairing differs from the Weil form. In an elementary Schur representation, an electric dressing followed by one magnetic insertion realizes the full positive single-prime reference, including every negative prime-power coefficient and its compulsory positive contact. A single quantization parameter works for all primes, and the insertion is justified by a graph-domain approximation. We derive restrictions on higher magnetic powers and exclude a constant-vector coherent gluing of two primes through its unwanted mixed translation atoms. We then prove that the resulting architecture cannot be completed at all: an elementary interference bound, applied with explicit test functions, shows that no coherent source has the positive gamma energy and the summed positive prime references as its two channel norms, at the first intervals where primes act and at every sufficiently long interval. Recording the target in its spectral form explains why, and identifies the mechanism that remains. We then exhibit it: a mirror of the positive prime reference, with the same prime atoms of the opposite sign and a strictly smaller contact, presents the target as a difference in which the remaining source carries no prime translation, and turns the criterion on each interval into a single domination, equivalently a compression. We then record what the two signed poles are: in the coordinates given by the evaluations at $s = 0$ and $s = 1$ the pole form is off-diagonal, a hyperbolic pairing of signature $(1, 1)$ whose null lines are the two evaluations, and every source for the localized form carries a unitary involution exchanging them, which for a compatible realization is the functional equation acting on the zeros. We then show that the pole term needs no mechanism of its own: splitting it by parity puts its positive half on the source side and its rank-one negative half on the subtracted side, and presents the target as a compression of the gamma energy in which both sides are positive term by term and the dominating side has an explicit source. That presentation assumes no unproved inequality and reduces the construction problem to exhibiting one critical contraction. We also settle the prime-free inequality outside a compact window, which is no longer a prerequisite for that problem, and characterise the source states reachable by one magnetic insertion: they are exactly the functions analytic past the unit circle, and the mirror prime reference is among them. Further exact tests cover interacting sphere states, a conformal electric sector, gauge transfer, and finite rational feedback. These results select and exclude source constructions; they do not establish Weil positivity or a field-theoretic reduction of the Riemann hypothesis.

1 Introduction

The Weil criterion expresses the Riemann hypothesis as positivity of an explicit quadratic form on compactly supported test functions. An inverse bulk approach asks for a quantum system, a boundary or defect preparation, and an independently positive physical pairing whose pullback is exactly that form. Interactions are compatible with this objective: a preparation linear in the input produces a quadratic state norm even when the underlying dynamics is interacting. What must be derived is the entire pairing, with its physical adjoint and normalization.

Here “boundary” means the locus or observable sector through which the arithmetic input enters the system. It may be a spatial boundary, a line defect, an interface, a gluing cut, or a module for a protected operator algebra. This flexibility permits the use of sphere and Schur quantization, where explicit representations and positive pairings are available [5, 6, 7]. Conformal symmetry can constrain a chosen defect, but does not identify arbitrary deformations or source maps. Pure four-dimensional Yang–Mills is also a possible long-term construction target; no identification with it is made in this paper.

1.1 Principal result and its limitation

The most concrete arithmetic result concerns the elementary Schur representation. For a fixed quantization parameter $0 < q < 1$, we construct an electric dressing and one magnetic insertion whose norm on Laurent polynomials h is

$$\|\Phi_{q;r,d}[h]\|^2 = \int_{S^1} |h(z)|^2 \frac{dr(1+r)}{1-r} \frac{|1-z|^2}{|1-rz|^2} \frac{d\theta}{2\pi}, \quad z = e^{i\theta}, \quad 0 < r < 1, \quad (1.1)$$

with $d > 0$. Its Fourier coefficients are

$$\frac{dr(1+r)}{1-r} \frac{|1-z|^2}{|1-rz|^2} = \frac{2dr}{1-r} - d \sum_{n \geq 1} r^n (z^n + z^{-n}). \quad (1.2)$$

Thus $r = p^{-1/2}$ and $d = \log p$ give every required single-prime repetition coefficient. The arithmetic parameters enter the preparation; q is held fixed. Theorem 6.2 proves this identity and the domain statement needed for its magnetic insertion. An explicit unitary decomposition then relates the circle calculation to the real logarithmic coordinate of the Weil form.

The positive contact $2dr/(1-r)$ is part of the result. This is a Schur realization of the positive prime reference already present in the project's conservative return construction. Neither the reference form nor the geometric Fourier identity is claimed as new. The additional content is the specified operator preparation, its source normalization, its graph-domain justification, and its compatibility with one fixed Schur parameter. A physical interface implementing the continuous-input lift has not been constructed.

1.2 Other results and the remaining question

The Gaussian real boundary module identifies the archimedean quarter-shift and oscillator levels, but its natural state-smearing norm has a different spectral weight. In the interacting $T[SU(2)]$ sphere model we evaluate an infinite family of charged-state norms and show that bare powers do not have geometric return ratios. A natural electric preparation in the conformal $SU(2)$ theory with four hypermultiplets also fails, as does the undressed elementary Schur return prescription. These are exclusions of specified preparations, not of the theories.

Two further results govern the new dressed source. Its finite magnetic domain does not permit unrestricted use of higher-power identities. Moreover, gluing the prime filters through fixed output vectors produces an atom at $\log(3/2)$ unless those vectors are orthogonal (Proposition 7.2). Orthogonality restores the independent positive contacts. A successful joint construction must therefore change the preparation or the gluing operation in a substantive way.

The sharpest result of this version says how far that change must go. Section 7.4 records the elementary bound $\|(G+S)f\|^2 \geq (\|Gf\| - \|Sf\|)^2$ and applies it to the two positive channels that the dressed construction produces, the gamma energy and the summed prime references. The bound fails at explicit test functions: at $L = 5/4$ the indicator of the interval already violates it, and it fails by a factor tending to infinity as L grows. Hence no coherent source has those two forms as its channel norms (Theorem 7.5), even though each is realized exactly and every prime coefficient matches. The contact and the pole term have to come from a compression or a projection inside one space, not from an additional positive channel (Proposition 7.3). Section 2.4 records the spectral form of the target, which explains the mechanism: the quantity to be matched is smaller than each term being matched by several orders of magnitude.

Section 7.5 then supplies such a presentation. Beside the positive prime reference there is a mirror reference (7.14) carrying the same prime atoms with the opposite sign and a strictly smaller

contact $2dr/(1+r)$. Its atoms cancel the Weil prime term outright, so the target becomes a difference (Theorem 7.8) in which the remaining form A_L has no prime translation at all, and the Weil criterion on I_L becomes one domination statement, equivalently a compression (Corollary 7.9). The prime-free form is not marginal where Q_L is, which is what makes it a construction target rather than a restatement.

Two results of this version simplify that target and withdraw a claim of the previous one. Section 6.5 characterises the states the elementary magnetic insertion can reach: exactly the functions analytic on a disk of radius exceeding one. The factor the operator inserts therefore constrains nothing, the mirror reference is realized by one dressing at the same fixed q (Corollary 7.10), and producing a prescribed weight in a positive sector is not by itself evidence of arithmetic structure in the representation. Section 7.6 then splits the pole form by parity and presents the target as $Q_L = K_+ - T_L$, with K_+ the gamma energy together with a constant and one rank-one term — a norm, with an explicit source — and T_L the negative contact, the odd half of the pole form and every mirror reference, positive term by term. Weil positivity on I_L becomes the single domination $T_L \preceq K_+$. Neither side needs an unproved inequality, so the prime-free inequality is not a prerequisite for the construction problem; and by Corollary 2.2 the pole term never required an indefinite-metric sector, which the previous version asserted in Section 8.2 and which is withdrawn here.

The background and normalization are developed in Section 2, independently of any companion manuscript. Section 3 states the conditional construction-to-RH implication and a general source-regularity restriction. Sections 4–7 contain the main calculations. Section 8 formulates the next complete fitting problem. The appendices retain positive gamma refinement and earlier gauge and rational-feedback tests, with proofs sufficient to use them without consulting the research notes.

The classical Weil criterion, gamma identities, and established representation formulas are explicitly attributed. The comparison calculations are working results, not a claim of literature priority. In particular, an exact identity between local phase derivatives and the prime terms is not a positivity theorem. No candidate in this draft matches the complete first-prime Weil form, and no all-support result follows from the local norm in (1.1).

Contents

1	Introduction	2
1.1	Principal result and its limitation	3
1.2	Other results and the remaining question	3
2	The complete Weil form in logarithmic coordinates	7
2.1	Conventions and the classical criterion	7
2.2	Positive gamma energy and the fixed contact	7
2.3	The signed poles and the localized target	8
2.4	The spectral form of the target	9
2.5	Parity and the hyperbolic pole pair	10
3	What an independently positive realization requires	11
3.1	State norms, constructive hypotheses, and linear sources	11
3.2	Protected pairings and their adjoints	12
3.3	A necessary source-regularity condition	12
4	Sphere quantization and the archimedean data	13
4.1	The real Gaussian boundary module	13
4.2	The oscillator tower and the distinction between a phase and a norm	14
4.3	An interacting benchmark: charged states in $T[SU(2)]$	14
5	Schur pairings before electric dressing	15
5.1	The conformal electric benchmark	15
5.2	Local difference factors and the full spherical vector	15
5.3	The common-state RG formulation and the undressed test	16
6	Dressed Schur returns and the exact positive prime norm	17
6.1	The elementary representation and electric preparation	17
6.2	One magnetic insertion and its graph domain	17
6.3	The full norm, including its contact	18
6.4	An explicit lift to continuous arithmetic inputs	19
6.5	Which output states the insertion can reach	19
7	Magnetic domains and joint-prime gluing	20
7.1	What higher magnetic powers would require	20
7.2	A two-prime obstruction for fixed output vectors	21
7.3	The full comparison, with no omitted contact or pole	22
7.4	Interference is necessary, and the two-channel architecture fails	22
7.5	The mirror reference and the subtraction form	24
7.6	The target as a compression of the gamma energy	27
7.7	The target as a jump form, and a necessary condition	29
8	The next complete matching problem	30
8.1	Relation to spectral realizations of the zeros	30
8.2	What the exclusion leaves	30
8.3	A finite-prime phase identity is useful background	32
8.4	A precise open problem	32
8.5	Scope of the working manuscript	33
A	A positive gamma action and its integer refinement	34

B Gauge transfer and two first-prime source tests	35
B.1 Winding a holonomy is different from transfer evolution	35
B.2 A positive disk system for masses and returns	36
B.3 An orthogonal first-prime branch has an unavoidable cusp	37
B.4 A coherent repair cancels the cusp and still fails the contact	37
C A finite rational-feedback obstruction	38
D Checks, provenance and mathematical status	38
References	41

2 The complete Weil form in logarithmic coordinates

2.1 Conventions and the classical criterion

All Hilbert inner products are conjugate-linear in the first argument. For $L > 0$, let $I_L = (-L/2, L/2)$ and let E_L denote extension by zero from I_L to $\mathbb{R}$. Unless another domain is specified, f belongs to $C_c^\infty(I_L; \mathbb{C})$ and $F = E_L f$. Our Fourier and translation conventions are

$$\widehat{F}(\tau) = \int_{\mathbb{R}} e^{-i\tau x} F(x) dx, \quad U_d F(x) = F(x - d), \quad \|F\|^2 = \frac{1}{2\pi} \int_{\mathbb{R}} |\widehat{F}(\tau)|^2 d\tau. \quad (2.1)$$

Write $\tilde{F}(x) = \overline{F(-x)}$ and $g = F * \tilde{F}$. Then $g(0) = \|F\|^2$, $g(-r) = \overline{g(r)}$, and $g(-r) = \langle F, U_r F \rangle$. In particular, the correlations vanish for $|r| \geq L$.

For $r > 0$ set

$$n_\gamma(r) = \frac{e^{-r/2}}{1 - e^{-2r}} = \sum_{n \geq 0} e^{-a_n r}, \quad a_n = 2n + \frac{1}{2}. \quad (2.2)$$

One normalization of the additive Weil functional is

$$\begin{aligned} \mathcal{W}(g) &= \int_{\mathbb{R}} 2 \cosh(x/2) g(x) dx - \sum_{p, m \geq 1} (\log p) p^{-m/2} \{g(m \log p) + g(-m \log p)\} \\ &\quad - (\log(4\pi) + \gamma_E) g(0) \\ &\quad - \int_0^\infty [g(r) + g(-r) - 2e^{-r/2} g(0)] n_\gamma(r) dr. \end{aligned} \quad (2.3)$$

Here p runs over rational primes and γ_E is Euler's constant. The integrand at the origin has a finite limit, and the prime sum is finite for compact support. This normalization can be found in Suzuki's account of the Weil form [1, Section 1.1], whose unconditional results on the lowest eigenvalue of the localized form are recorded in Remark 7.12; for its geometric and trace interpretation see [2].

The Riemann hypothesis (RH) asserts that every nontrivial zero of the meromorphic continuation of $\zeta(s) = \sum_{n \geq 1} n^{-s}$ has real part $1/2$. We use the following classical theorem as the number-theoretic input; we do not reprove the explicit formula or the Weil criterion.

Theorem 2.1 (Classical Weil positivity criterion). *RH holds if and only if $\mathcal{W}(F * \tilde{F}) \geq 0$ for every complex $F \in C_c^\infty(\mathbb{R})$.*

Every compact support is contained in some I_L . Consequently it is sufficient, and necessary, to establish this positivity for every $L > 0$ and every test function in I_L .

2.2 Positive gamma energy and the fixed contact

Define

$$\begin{aligned} K[F] &= \frac{1}{2} \int_{\mathbb{R}} \int_{\mathbb{R}} |F(x) - F(y)|^2 n_\gamma(|x - y|) dx dy \\ &= \int_0^\infty [2g(0) - g(r) - g(-r)] n_\gamma(r) dr. \end{aligned} \quad (2.4)$$

This is nonnegative without any arithmetic hypothesis. Expanding (2.2) and applying Plancherel gives

$$K[F] = \frac{1}{2\pi} \int_{\mathbb{R}} b(\tau^2) |\widehat{F}(\tau)|^2 d\tau, \quad (2.5)$$

$$b(\tau^2) = \sum_{n \geq 0} \frac{2}{a_n} \frac{\tau^2}{a_n^2 + \tau^2} = \operatorname{Re} \psi\left(\frac{1}{4} + i\tau/2\right) - \psi\left(\frac{1}{4}\right), \quad (2.6)$$

where $\psi = \Gamma'/\Gamma$. Indeed, $2 \int_0^\infty (1 - \cos(\tau r)) e^{-ar} dr = (2/a)\tau^2/(a^2 + \tau^2)$; nonnegativity justifies interchange with the sum. The second identity in (2.6) is the digamma partial-fraction expansion [3].

The archimedean part of (2.3) equals $K[F] + w_0 \|F\|^2$, where

$$\begin{aligned} w_0 &= -\log(4\pi) - \gamma_E - 2 \int_0^\infty (1 - e^{-r/2}) n_\gamma(r) dr \\ &= \psi(\tfrac{1}{4}) - \log \pi = -\gamma_E - \frac{\pi}{2} - 3 \log 2 - \log \pi < 0. \end{aligned} \quad (2.7)$$

To verify the constant, the integral in the first line is $\sum_{n \geq 0} (a_n^{-1} - (2n+1)^{-1}) = \frac{1}{2}[\psi(1/2) - \psi(1/4)]$. The constant is therefore fixed by the arithmetic normalization. It is not a parameter available to adjust a positive source.

2.3 The signed poles and the localized target

The first term of (2.3) has kernel $2 \cosh((x-y)/2)$. Using the hyperbolic addition formula gives

$$P_L[f] = 2 \left| \int_{I_L} f(x) \cosh(x/2) dx \right|^2 - 2 \left| \int_{I_L} f(x) \sinh(x/2) dx \right|^2. \quad (2.8)$$

It is a rank-two Hermitian form, with one positive and one negative direction for $L > 0$. Its pointwise positive kernel does not make it a positive quadratic form.

The exact target for this paper is

$$\begin{aligned} Q_L[f] := \mathcal{W}(F * \tilde{F}) &= K[F] + w_0 \|f\|^2 + P_L[f] \\ &\quad - 2 \sum_{m \log p < L} (\log p) p^{-m/2} \operatorname{Re} \langle F, U_{m \log p} F \rangle. \end{aligned} \quad (2.9)$$

Its polarization is denoted by $Q_L(f, g)$. At equality $m \log p = L$ the compressed correlation is zero. The coefficient is the primitive length $\log p$, not the repeated length $m \log p$.

On a fixed interval the prime and pole terms define bounded operators. The gamma multiplier satisfies

$$b(\tau^2) = \log |\tau| - \log 2 - \psi(\tfrac{1}{4}) + O(|\tau|^{-2}) \quad (|\tau| \rightarrow \infty). \quad (2.10)$$

The natural energy domain is

$$\mathcal{D}_{\gamma, L} = \left\{ f \in L^2(I_L) : \int_{\mathbb{R}} \log(2 + |\tau|) |\widehat{E_L f}(\tau)|^2 d\tau < \infty \right\}. \quad (2.11)$$

The positive gamma energy is closed on this domain; it is the restriction of the closed nonnegative Fourier-multiplier form to the closed subspace of functions supported in $\bar{I}_L$. The bounded other terms give a closed semibounded form on the same domain. All matching statements below are first required on smooth compactly supported inputs. The Weil criterion itself needs only that test space.

Away from the diagonal the distribution kernel of K is $-n_\gamma(|x-y|)$, which is smooth whenever $x \neq y$. The prime contribution instead has atoms at $x-y = \pm m \log p$. This distinction will provide exact tests for candidate preparations.

Three small intervals already distinguish several issues:

L	Active prime powers	Test
1	2	First contact, pole and translation identity
5/4	2, 3	Joint source and mixed returns
3/2	2, 3, 4	First nontrivial repetition of 2

A calculation on any one interval is preliminary. It does not establish the all-support statement of Theorem 2.1.

2.4 The spectral form of the target

The target (2.9) has a classical spectral description. We record it because it governs what a realization has to produce, and because it controls the size of the quantity being matched.

For $g = F * \tilde{F}$ with $F \in C_c^\infty(\mathbb{R})$, the function

$$G(\tau) = \int_{\mathbb{R}} g(x) e^{-i\tau x} dx = \widehat{F}(\tau) \overline{\widehat{F}(\bar{\tau})} \quad (2.12)$$

is entire of exponential type and agrees with $|\widehat{F}(\tau)|^2$ for real τ . Write ρ for the nontrivial zeros of ζ with multiplicity and put $\gamma_\rho = -i(\rho - \frac{1}{2})$. In the normalization (2.3), Weil's explicit formula states that

$$\mathcal{W}(g) = \sum_{\rho} G(\gamma_\rho), \quad (2.13)$$

the sum taken in the symmetric order $|\operatorname{Im} \rho| \leq T$, $T \rightarrow \infty$ [1, 2]. The first term of (2.3) is $G(i/2) + G(-i/2)$, the contribution of the two poles of $\zeta(s)$ at $s = 0$ and $s = 1$; the remaining terms are the archimedean and finite local factors. We use (2.13) as classical input and do not reprove it. Appendix D records a direct check that the normalization (2.3) used here reproduces it.

Corollary 2.2 (Spectral form of the localized target). *For $f \in C_c^\infty(I_L)$ and $F = E_L f$,*

$$Q_L[f] = \sum_{\rho} \widehat{F}(\gamma_\rho) \overline{\widehat{F}(\bar{\gamma}_\rho)}. \quad (2.14)$$

Under RH every γ_ρ is real and each summand is $|\widehat{F}(\gamma_\rho)|^2 \geq 0$. A zero off the critical line pairs with the zero $1 - \bar{\rho}$ to contribute $2 \operatorname{Re}[\widehat{F}(\gamma) \overline{\widehat{F}(\bar{\gamma})}]$, which has no fixed sign.

Three consequences organize what follows.

First, (2.14) determines a realization that is compatible in L . Suppose Φ is defined on $C_c^\infty(\mathbb{R})$ and satisfies $\|\Phi(f)\|^2 = Q_L[f]$ whenever $\operatorname{supp} f \subset I_L$. Then RH holds, and $\Phi(f) \mapsto (\widehat{F}(\gamma_\rho))_\rho$ extends to a unitary map from the closure of the range of Φ onto the closed span of those vectors in ℓ^2 of the zero multiset. The Hilbert space and the source of a compatible realization are therefore not free data: up to that unitary they are the zeros and the evaluation map.

Second, this fixes the shape of the answer without deciding the objective of this manuscript. That some positive Hilbert space carries this spectrum is equivalent to RH and cannot be arranged by fiat. The proposal here is more specific: that an independently motivated family of positive pairings — protected line and defect algebras with physically defined twisted traces — contains such a source, so that construction of a named theory with specified observables would supply it. Nothing in (2.14) decides whether that is so.

Third, (2.14) is a warning about method. Its right side is generally far smaller than the individual terms of (2.9). For the smooth input used in Appendix D on I_4 , the gamma, contact, pole and prime terms of (2.9) are 3.40, -8.06 , 9.11 and -4.46 , while their sum is 1.4×10^{-7} . A candidate is therefore not usefully tested by matching those terms one at a time. The cancellation between them is the content of the problem, and Section 7.4 turns that remark into an exact obstruction.

2.5 Parity and the hyperbolic pole pair

The rank-two form (2.8) is usually read through its positive and negative directions. A different basis makes its structure, and the constraint it places on a realization, immediate.

Lemma 2.3 (The pole form is a hyperbolic pair). *For $f \in C_c^\infty(I_L)$ and $F = E_L f$ put*

$$u[f] = \widehat{F}(i/2) = \int f(x) e^{x/2} dx, \quad v[f] = \widehat{F}(-i/2) = \int f(x) e^{-x/2} dx, \quad (2.15)$$

the evaluations at the two poles of ζ , at $s = 1$ and $s = 0$. Then $\langle \cosh(x/2), f \rangle = (u + v)/2$, $\langle \sinh(x/2), f \rangle = (u - v)/2$, and

$$P_L[f] = 2 \operatorname{Re} (u[f] \overline{v[f]}). \quad (2.16)$$

In the coordinates (u, v) the form is therefore off-diagonal, with matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$: it is the hyperbolic form of signature $(1, 1)$, and its two null lines are $\{u = 0\}$ and $\{v = 0\}$.

Proof. $e^{\pm x/2} = \cosh(x/2) \pm \sinh(x/2)$ gives the two moment identities, and $2| \frac{u+v}{2} |^2 - 2| \frac{u-v}{2} |^2 = 2 \operatorname{Re}(u\bar{v})$. The matrix statement is (2.16) read as a Hermitian form in (u, v) ; the diagonal entries vanish, so each coordinate axis is null and the signature is $(1, 1)$. $\square$

The negative direction of P_L is thus not attached to either pole: it is the antisymmetric combination, and the indefiniteness lies in the pairing between the two poles rather than in one of them.

The same two combinations are the parity components. Let $(Rf)(x) = f(-x)$. Every kernel in (2.9) depends on $x - y$ through an even function, so $Q_L(Rf, Rg) = Q_L(f, g)$, and the same holds for any form obtained from it by changing the constant. Both forms therefore decouple over the eigenspaces of R , and since $\cosh(x/2)$ is even and $\sinh(x/2)$ odd,

$$P_L|_{\text{even}} = +2 |\langle \cosh(x/2), f \rangle|^2, \quad P_L|_{\text{odd}} = -2 |\langle \sinh(x/2), f \rangle|^2. \quad (2.17)$$

The indefiniteness of the pole form is an odd-sector phenomenon and its positive direction is entirely even.

Proposition 2.4 (Every source carries the involution). *Let Φ be defined on $C_c^\infty(I_L)$ with values in a Hilbert space and $\|\Phi(f)\|^2 = Q_L[f]$. Then there is a unitary J on the closure of the span of the range of Φ with $J\Phi(f) = \Phi(Rf)$ and $J^2 = \operatorname{id}$. The positive direction of (2.17) lies in the $+1$ eigenspace of J and the negative direction in the -1 eigenspace. If moreover Φ is compatible in L in the sense of Corollary 2.2, then under the unitary of that corollary J is the involution $\rho \mapsto 1 - \rho$ of the zero multiset.*

Proof. Polarization gives $Q_L(Rf, Rg) = Q_L(f, g)$, so $\langle \Phi(Rf), \Phi(Rg) \rangle = \langle \Phi(f), \Phi(g) \rangle$ and $\Phi(f) \mapsto \Phi(Rf)$ extends from the span of the range to an isometry, surjective because $R^2 = \operatorname{id}$; call it J . The eigenspace statement is (2.17). For the last statement, $\widehat{E_L Rf}(\tau) = \widehat{E_L f}(-\tau)$, and the multiset $\{\gamma_\rho\}$ is invariant under $\gamma \mapsto -\gamma$ because the functional equation pairs ρ with $1 - \rho$; so on the coordinates of Corollary 2.2 the map J permutes γ_ρ with $\gamma_{1-\rho} = -\gamma_\rho$. $\square$

The two poles $s = 0$ and $s = 1$ are exchanged by the same involution, being the two points the functional equation adds to the zero set. Lemma 2.3 and Proposition 2.4 are used in Section 8.2 to state what a realization of the pole term has to contain.

3 What an independently positive realization requires

3.1 State norms, constructive hypotheses, and linear sources

Definition 3.1 (Exact positive realization). An exact positive realization of the Weil form consists of a specified model M with a positive Hilbert space $\mathcal{H}_M$ and complex-linear maps $\Phi_{M,L} : C_c^\infty(I_L) \rightarrow \mathcal{H}_M$ such that

$$\langle \Phi_{M,L}(f), \Phi_{M,L}(g) \rangle = Q_L(f, g) \quad (L > 0). \quad (3.1)$$

The model, its physical inner product, source rule and normalization must be specified independently of an assumed positive Weil form. Construction of the model may be an explicit hypothesis; the identity (3.1) must be derived from the specified data.

One usual form is $\Phi_{M,L}(f) = \mathcal{O}_{M,L}(f)\Omega_M$. The observable $\mathcal{O}_{M,L}(f)$ need not be a positive operator. The positivity argument is

$$\langle \mathcal{O}(f)\Omega, \mathcal{O}(g)\Omega \rangle, \quad \|\mathcal{O}(f)\Omega\|^2 \geq 0, \quad (3.2)$$

provided these vectors exist in the positive physical Hilbert space. Writing this formally as an expectation of $\mathcal{O}(f)^\dagger \mathcal{O}(g)$ is justified only on its operator or quadratic-form domain. Linearity of the state map supplies quadratic dependence on f ; the interaction strength of the theory does not affect this argument.

A Euclidean formulation may use a preparation $A_L(f)$ at positive Euclidean time and reflection Θ :

$$B_{M,L}(f, g) = \langle (\Theta A_L(f)) A_L(g) \rangle_M. \quad (3.3)$$

Reflection positivity supplies its positive norm interpretation; a full relativistic reconstruction requires the other Osterwalder–Schrader hypotheses [10]. The arithmetic variable x in (2.9) is not automatically Euclidean time.

An alternative classical construction is a nonnegative joint energy $\mathcal{E}_M(\Phi)$ with boundary trace f , for which $Q_L[f] = \inf_{\text{Tr } \Phi=f} \mathcal{E}_M(\Phi)$. Positivity only in the auxiliary Hessian is insufficient: $w^2 - 2fw$ is coercive in w and has minimum $-f^2$. For nonlinear energies, the assertion that the entire minimized boundary dependence is quadratic is itself a matching requirement.

Proposition 3.2 (Conditional construction-to-RH implication). *If the construction hypotheses for a specified model imply the positive Hilbert space and domains in Definition 3.1, and the complete matching identity is derived under those hypotheses, then construction of that model with those observables implies RH. It is also sufficient to have independently positive regulated pairings $B_{\Lambda,L}$ such that, for every fixed pair of test functions and L ,*

$$\lim_{\Lambda \rightarrow \infty} B_{\Lambda,L}(f, g) = Q_L(f, g). \quad (3.4)$$

Proof. The first assertion follows by taking $g = f$ in (3.1) and applying Theorem 2.1. In the second assertion the finite limit of nonnegative real numbers $B_{\Lambda,L}(f, f)$ is nonnegative. The same criterion applies. No rate uniform in all inputs is required. $\square$

The proposition is a logical consequence of the classical criterion, not the missing reduction theorem. We have not specified a model for which its full premise is verified. An additive subtraction from a regulated norm need not remain positive. Likewise, defining a covariance to equal Q_L , or setting $\Phi = Q_L^{1/2}$ before positivity is known, would reinsert the original problem into the construction hypothesis.

A common source rule compatible under enlargement of I_L is not a convenience. Separate positive realizations for every L would suffice logically for RH, but a realization at one fixed interval is not by itself informative: as soon as $Q_L \geq 0$ holds, $\Phi(f) = Q_L^{1/2}f$ into $L^2(I_L)$ already satisfies Definition 3.1 on that interval. The content of a proposed realization is carried entirely by the uniformity of its source rule in L , and by Corollary 2.2 that uniform object is the zero measure. Short intervals are therefore places to test and discard a specified candidate, not places where a partial result accumulates.

Remark 3.3 (Stationary data at one interval). The same conclusion holds inside the narrower class of translation-covariant sources. If $Q_L \geq 0$ and $\varphi \in C_c^\infty$ has small support, then $r \mapsto Q_L(\varphi, U_r \varphi)$ is a continuous positive definite function on an interval, because $\sum_{j,k} \bar{c}_j c_k Q_L(U_{r_j} \varphi, U_{r_k} \varphi) = Q_L[\sum_k c_k U_{r_k} \varphi] \geq 0$ whenever the translates stay inside I_L . Krein's extension theorem [20, 21] then supplies a finite positive measure whose Fourier transform agrees with it there, hence stationary positive data on that scale. Positivity on one interval already yields such data; only its compatibility as L grows carries arithmetic information. This remark is not used in any proof below.

A mass gap is not needed for the elementary norm implication. It must be imposed only if a chosen construction or observable limit requires it. An extension of pure Yang–Mills by arithmetic defects or extra fields would create construction obligations beyond the pure Yang–Mills existence problem [12].

3.2 Protected pairings and their adjoints

Sphere and Schur quantization motivate Hilbert completions of protected algebras. The basic data can be described as a complex algebra $\mathcal{A}$, an antilinear automorphism ρ , and a twisted trace T with positive Hermitian form

$$H(a, b) = T(\rho(a)b), \quad T(ab) = T(\rho^2(b)a). \quad (3.5)$$

One quotients null vectors when necessary before completion. On the algebraic core the right and left actions satisfy $R_a^\dagger = L_{\rho(a)}$, since $H(ba, c) = T(\rho(b)\rho(a)c) = H(b, \rho(a)c)$. This does not identify ρ with an arbitrary ordinary adjoint on a single copy of the algebra, nor establish every unbounded closure. The representation formulas used below come from [5, 6, 7]. Classification results for related positive forms give independent mathematical support in specified algebraic families [8, 9]; they do not construct every correlator of an interacting continuum field theory.

The intended conditional program allows an open but credible construction problem. Explicit positive representations, stable regulated models, protected localization formulas and compatible dual descriptions provide different kinds of evidence. The source domain and matching identity still have to be stated for the particular model. No general Schur construction conjecture is promoted to a theorem in the calculations below.

3.3 A necessary source-regularity condition

Proposition 3.4 (Finite-norm orbit smearing cannot match). *Fix $L > 0$. A source of the form $\Phi(f) = \int_{I_L} f(x) V_x \Omega dx$, with V_x unitary, $\Omega \in \mathcal{H}$, and a well-defined strongly measurable orbit, cannot have norm $Q_L[f]$ for all smooth test functions. The same holds for a finite direct sum of such sources.*

Proof. The estimate $\|\Phi(f)\| \leq \sqrt{L}\|\Omega\|\|f\|$ makes the proposed pairing bounded in the input L^2 norm. Let $0 \neq \chi \in C_c^\infty(I_L)$ and $f_N(x) = \chi(x)e^{iNx}$. Equation (2.10) and rapid decay of $\hat{\chi}$ give

$$\frac{Q_L[f_N]}{\log N} \longrightarrow \|\chi\|^2. \quad (3.6)$$

For completeness, after the change $\tau = N + u$ the integrand of the gamma term divided by $\log N$ converges pointwise to $|\widehat{\chi}(u)|^2$. The bound $b((N + u)^2) \leq C[1 + \log(2 + N) + \log(2 + |u|)]$ for $N \geq 2$ permits dominated convergence. All other terms in (2.9) are bounded uniformly in N on this interval. This contradicts boundedness of the proposed norm. $\square$

An exact source must be unbounded relative to the input L^2 norm. An infinite family of insertions or a distributional field with finite output on smooth tests is compatible with this requirement. The proposition makes no assertion of global closability on the union of all intervals, and imposes no obstruction on QFT as such.

4 Sphere quantization and the archimedean data

4.1 The real Gaussian boundary module

The real boundary module of the free hypermultiplet provides a Schrödinger representation [5, Sections 7.2–7.3]. We use oscillator units and the explicitly normalized data

$$\mathcal{H}_{\mathbb{R}} = L^2(\mathbb{R}, dy), \quad X = y = X^\dagger, \quad P = \partial_y = -P^\dagger, \quad \Omega(y) = \pi^{-1/4} e^{-y^2/2}. \quad (4.1)$$

The unitary dilation group is $D_r \psi(y) = e^{r/2} \psi(e^r y)$. Direct integration gives

$$\langle \Omega, D_r \Omega \rangle = (\cosh r)^{-1/2}. \quad (4.2)$$

The real, even Gaussian state fixes a particular Mellin weight; no adjustable gamma shift is introduced.

Proposition 4.1 (Gaussian Mellin normalization). *On even states, define $(\mathcal{U}\psi)(u) = \sqrt{2}e^{u/2}\psi(e^u)$ and let $\mathcal{M}$ be its unitary Fourier transform with factor $(2\pi)^{-1/2}$. Then*

$$(\mathcal{M}\Omega)(\tau) = 2^{-3/4-i\tau/2} \pi^{-3/4} \Gamma(\tfrac{1}{4} - i\tau/2), \quad (4.3)$$

$$\left\| \int_{\mathbb{R}} f(r) D_r \Omega \, dr \right\|^2 = \int_{\mathbb{R}} \rho_G(\tau) |\widehat{f}(\tau)|^2 \, d\tau, \quad \rho_G(\tau) = \frac{|\Gamma(\tfrac{1}{4} + i\tau/2)|^2}{2^{3/2} \pi^{3/2}}. \quad (4.4)$$

The density ρ_G integrates to one.

Proof. Evenness makes $\mathcal{U}$ unitary by the substitution $y = e^u$. The transform of the Gaussian reduces to $\sqrt{2}\pi^{-1/4}(2\pi)^{-1/2} \int_0^\infty y^{-1/2-i\tau} e^{-y^2/2} dy$. The substitution $y^2/2 = t$ evaluates this integral and proves (4.3). Dilations become translations in u , so spectral diagonalization proves (4.4); the possible sign of the Fourier argument is immaterial because ρ_G is even. Plancherel and $\|\Omega\| = 1$ give the last assertion. $\square$

This is an exact positive pairing associated with the quarter-shift. It is not Q_L : its kernel (4.2) is smooth and its spectral density decays exponentially, while (2.10) grows logarithmically. It also falls under Proposition 3.4. Thus this natural source is a normalization control rather than the desired full preparation.

4.2 The oscillator tower and the distinction between a phase and a norm

The positive oscillator $H_{\text{osc}} = \frac{1}{2}(-\partial_y^2 + y^2)$ has, on its even subspace, the spectrum $a_n = 2n + 1/2$. Consequently

$$b(\tau^2) = 2 \text{Tr}_{\mathcal{H}_{\text{even}}} \left[H_{\text{osc}}^{-1} \frac{\tau^2}{H_{\text{osc}}^2 + \tau^2} \right]. \quad (4.5)$$

Each fixed- τ trace converges, since its summands are $O(n^{-3})$. An explicit positive source into $\bigoplus_{n \geq 0} L^2(\mathbb{R})$ is

$$(\mathcal{A}_n F)^\sim(\tau) = \sqrt{\frac{2}{a_n}} \frac{i\tau}{a_n + i\tau} \widehat{F}(\tau), \quad \sum_{n \geq 0} \|\mathcal{A}_n F\|^2 = K[F]. \quad (4.6)$$

Equivalently one may use diagonal Hilbert–Schmidt operators in the oscillator basis. The source is finite on the domain of K and is closed there as a map with its graph norm. Equation (4.5) identifies the required masses with a standard Hamiltonian in a named module. It does not derive the resolvent-weighted source (4.6) as a local insertion or a gluing map of that QFT.

The corresponding unit-modulus archimedean phase is

$$S_\infty(\tau) = \pi^{-i\tau} \frac{\Gamma(\frac{1}{4} + i\tau/2)}{\Gamma(\frac{1}{4} - i\tau/2)}, \quad -i\overline{S_\infty} S'_\infty = b(\tau^2) + w_0. \quad (4.7)$$

Differentiating its logarithm proves the second identity. The scalar factor $\pi^{-i\tau}$ is essential for the contact. Although $|S_\infty| = 1$, its phase derivative at zero is $w_0 < 0$. Unitarity of a multiplication operator therefore does not establish positivity of this derivative. Local-factor phases and gamma splitting are classical; their semilocal use is developed, for example, in [4].

4.3 An interacting benchmark: charged states in $T[SU(2)]$

The $T[SU(2)]$ theory is the three-dimensional $U(1)$ theory with two hypermultiplets. Its sphere module admits a representation on twisted half-densities on $\mathbb{CP}^1$ [5, Section 3.2]. For real FI parameter t , use the coordinate-chart data

$$E = \partial_z, \quad \Omega_t(z) = (1 + |z|^2)^{-1+it}, \quad d\mu = d^2z/\pi. \quad (4.8)$$

This vector has unit norm. Multiplication by $\Gamma(1 - it)$ restores the partition-function factor $Z_t = \pi t / \sinh(\pi t)$, understood continuously as $Z_0 = 1$.

Proposition 4.2 (Charged-state Gram matrix). *The states in (4.8) satisfy*

$$E^n \Omega_t = (-1)^n (1 - it)_n \bar{z}^n (1 + |z|^2)^{-1+it-n}, \quad (4.9)$$

$$\langle E^m \Omega_t, E^n \Omega_t \rangle = \delta_{mn} N_n(t), \quad N_n(t) = \frac{(n!)^2}{(2n+1)!} \prod_{k=1}^n (k^2 + t^2). \quad (4.10)$$

Neither the vacuum returns nor the ratios of these norms supply a nonzero geometric repetition law for bare powers of a fixed multiple of E .

Proof. Repeated differentiation proves (4.9). Angular integration makes different powers orthogonal. With $u = |z|^2$, the remaining radial integral is $\int_0^\infty u^n (1+u)^{-2n-2} du = (n!)^2 / (2n+1)!$. Also $|(1-it)_n|^2 = \prod_{k=1}^n (k^2 + t^2)$. Thus $\langle \Omega_t, E^n \Omega_t \rangle = 0$ for $n > 0$, whereas

$$\frac{N_{n+1}(t)}{N_n(t)} = \frac{(n+1)^2((n+1)^2 + t^2)}{(2n+3)(2n+2)}. \quad (4.11)$$

The ratio at $n = 1$ exceeds that at $n = 0$ by $(19 + t^2)/30 > 0$. A nonzero constant rescaling of E multiplies both ratios by the same positive number and cannot make them equal. $\square$

These are norms in an interacting protected sector. The result excludes the proposed bare charged return, not mixed charged and neutral observables, vortex modules or interfaces. A source $\sum_n c_n(f) E^n \Omega_t$ has norm $\sum_n N_n(t) |c_n(f)|^2$ when the sum converges; an arithmetic application still needs an independent rule for the coefficients $c_n(f)$.

5 Schur pairings before electric dressing

5.1 The conformal electric benchmark

Consider the four-dimensional $\mathcal{N} = 2$ $SU(2)$ theory with four hypermultiplets, with $0 < q < 1$ and trivial flavour fugacities. Let

$$(a; Q)_N = \prod_{j=0}^{N-1} (1 - aQ^j), \quad (a; Q)_\infty = \prod_{j \geq 0} (1 - aQ^j). \quad (5.1)$$

The electric-sector spherical norm from [6, Section 5.1.5] specializes to the finite positive measure

$$\begin{aligned} d\nu_q(\theta) &= Z_q^{-1} \omega_q(\theta) \frac{d\theta}{2\pi}, \quad Z_q = \int_0^{2\pi} \omega_q(\theta) \frac{d\theta}{2\pi}, \\ \omega_q(\theta) &= 4 \sin^2 \theta \frac{|(q^2 e^{2i\theta}; q^2)_\infty|^4}{|(-q e^{i\theta}; q^2)_\infty|^{16}}. \end{aligned} \quad (5.2)$$

The Wilson operator acts as $W = 2 \cos \theta$. Normalizing by Z_q sets the vacuum norm to one; the overall physical normalization would also have to be fixed by a full matching theorem.

For the natural source $a_f(W) = \int f(x) e^{ixW} dx$, the entire pairing is

$$B_q(f, g) = \int_0^{2\pi} \overline{\widehat{f}(-2 \cos \theta)} \widehat{g}(-2 \cos \theta) d\nu_q(\theta). \quad (5.3)$$

Its kernel $\int e^{2ir \cos \theta} d\nu_q(\theta)$ is entire in r , because the spectral variable lies in $[-2, 2]$. It cannot produce separated prime atoms, and its bounded input norm contradicts Proposition 3.4. This conclusion concerns this electric preparation only.

5.2 Local difference factors and the full spherical vector

The elementary factor

$$\chi_q(v) = (-qv; q^2)_\infty^{-1} \quad \text{satisfies} \quad \frac{\chi_q(q^2 v)}{\chi_q(v)} = 1 + qv. \quad (5.4)$$

It occurs in the elementary q -Weyl example of Ambrosino–Gaiotto [7, Section 3.1.1]. The ratio follows by cancelling the products in (5.1). For a trial substitution $q = p^{-1/2}$ and $v = -e^{i\tau \log p}$, let $J_p(\tau) = 1 - p^{-1/2} e^{i\tau \log p}$. Then

$$2 \operatorname{Im} \partial_\tau \log J_p(\tau) = -2(\log p) \sum_{m \geq 1} p^{-m/2} \cos(m\tau \log p). \quad (5.5)$$

The series converges absolutely. The unshifted logarithm of χ_q instead has coefficients with a factor $(1 - q^{2m})^{-1}$. Thus the ratio in (5.4), rather than the factor itself, supplies the geometric coefficient.

This calculation has two immediate limitations. A phase derivative is not a state norm. In addition, varying q with p varies the quantization parameter, and is not a construction in one fixed theory. The dressed source in the next section addresses both points for the positive local reference. In the full massless four-flavour benchmark, the spherical vector has instead the complete ratio

$$\frac{\varphi_0(q^2 v)}{\varphi_0(v)} = \frac{(1 + qv)^8}{(1 - v^2)(1 - q^2 v^2)^2(1 - q^4 v^2)}. \quad (5.6)$$

All matter and vector factors are retained here. Extracting one numerator factor would change the model calculation. Complex holonomy shifts and changes of magnetic charge also require an explicit source map before they can represent translation in the real arithmetic coordinate.

5.3 The common-state RG formulation and the undressed test

The RG formulation [7, Sections 1–2] transports a UV line a to a framed expansion F_a acting on a common spectrum-generator state $\mathcal{S}$. For charge lattice Γ , the relevant quantum torus and pairing take the form

$$X_\gamma X_\eta = q^{\langle \gamma, \eta \rangle} X_{\gamma + \eta}, \quad I(a, b) = c_q \langle F_a \mathcal{S}, F_b \mathcal{S} \rangle_{\ell^2(\Gamma)}, \quad c_q = (q^2; q^2)_\infty^{2r_g} > 0. \quad (5.7)$$

Here $\langle \gamma, \eta \rangle$ is the integral charge pairing and r_g is the gauge rank. The same state enters both the dualization rule and the norm. General domain and construction claims in this formalism include conjectural input; only the explicitly defined elementary Hilbert-space calculations below are used as unconditional statements. An isometric RG description preserves the norm of a given source and cannot itself repair a mismatch.

On one charge ray, take the bilateral shift $Xe_n = e_{n+1}$ on $\ell^2(\mathbb{Z})$ and the vector

$$\mathcal{S} = \sum_{n \geq 0} c_n e_n, \quad c_n = \frac{(-q)^n}{(q^2; q^2)_n}, \quad Z = \sum_{n \geq 0} |c_n|^2 < \infty. \quad (5.8)$$

These are the Fourier coefficients of χ_q . The recurrence $(1 - q^{2n})c_n = -qc_{n-1}$ gives the complete norm identity

$$\|(1 + qX)\mathcal{S}\|^2 = (1 + q^2)Z + 2q \operatorname{Re} \langle \mathcal{S}, X\mathcal{S} \rangle = \sum_{n \geq 0} q^{4n} |c_n|^2. \quad (5.9)$$

Indeed each coefficient of $(1 + qX)\mathcal{S}$ is $q^{2n}c_n$, including the coefficient at $n = 0$.

The actual normalized overlaps do not give q^m . For $m \geq 1$,

$$(-1)^m \frac{\langle \mathcal{S}, X^m \mathcal{S} \rangle}{Z} = \frac{q^m}{Z} \sum_{n \geq 0} \frac{|c_n|^2}{\prod_{j=1}^m (1 - q^{2(n+j)})} > q^m. \quad (5.10)$$

Every summand is positive and each denominator is strictly less than one. Convergence follows from $(q^2; q^2)_n \rightarrow (q^2; q^2)_\infty > 0$. Thus a correct local difference equation does not by itself determine the desired physical return coefficient. This exact failure motivates changing the prepared state while retaining the positive Hilbert metric.

6 Dressed Schur returns and the exact positive prime norm

6.1 The elementary representation and electric preparation

Fix one real parameter $0 < q < 1$. This parameter will be the same for all primes. The elementary q -Weyl representation associated with SQED with one hypermultiplet has the following data [7, Section 3.1.1]. Let

$$\mathcal{H} = \ell^2(\mathbb{Z}) \otimes L^2(S^1, d\theta/(2\pi)), \quad \zeta = e^{i\theta}, \quad (6.1)$$

and write the magnetic charge as m . On the appropriate algebraic line-state domain the actions are

$$\begin{aligned} (v\psi)_m(\zeta) &= q^{-m}\zeta\psi_m(\zeta), \\ (u_-\psi)_m(\zeta) &= (1 + q^{-m-1}\zeta)\psi_{m+1}(q^{-1}\zeta), \\ (u_+\psi)_m(\zeta) &= \psi_{m-1}(q\zeta), \\ (\Omega_q)_m(\zeta) &= C_q\delta_{m0}\chi_q(\zeta), \quad C_q = (q^2; q^2)_\infty. \end{aligned} \quad (6.2)$$

The physical adjoints are inherited from the doubled representation; we do not use $u_- = u_+^\dagger$. Positivity in the calculation is the ordinary positive norm of (6.1). The elementary representation is a protected-sector control, not an identification with the conformal four-flavour theory of Section 5.

For $0 < a < 1$ define, on the zero-charge sector,

$$D_{q,a}(\zeta) = \frac{\sqrt{1-a^2}(-q\zeta; q^2)_\infty}{C_q(1+a\zeta)}. \quad (6.3)$$

The numerator is entire in ζ and the expression is analytic in a neighbourhood of the closed unit disk. Thus its Taylor series defines an electric-line completion, bounded on that sector. It is not asserted to be bounded on all magnetic sectors. The prepared state is explicitly

$$\eta_a = D_{q,a}(v)\Omega_q = \delta_{m0} \frac{\sqrt{1-a^2}}{1+a\zeta}, \quad \|\eta_a\| = 1. \quad (6.4)$$

Multiplication by $-v$ on this sector gives

$$\langle \eta_a, (-v)^k \eta_a \rangle = a^{|k|}, \quad k \in \mathbb{Z}. \quad (6.5)$$

Both assertions follow from the geometric coefficients $\sqrt{1-a^2}(-a)^n$, $n \geq 0$. Thus the descendant factors in (5.10) have been removed by changing the source, without changing the metric. The parameter a is independent of q . This infinite electric preparation is additional specified observable data; it has not been derived as a unique UV boundary condition.

6.2 One magnetic insertion and its graph domain

Lemma 6.1. *If $a = qr$ with $0 < r < 1$, the vector η_{qr} belongs to the graph closure of the algebraic negative magnetic action on its electric-line approximants. Its image is*

$$(u_-\eta_{qr})_m(\zeta) = \delta_{m,-1} \sqrt{1-q^2r^2} \frac{1+\zeta}{1+r\zeta}, \quad \|u_-\eta_{qr}\|^2 = \frac{2(1-q^2r^2)}{1+r}. \quad (6.6)$$

Every closed physical extension of that algebraic action contains this graph limit and has the displayed value.

Proof. The formula for the state follows by substituting (6.4) in (6.2). To justify that substitution as an operator action, let D_N be the Taylor polynomials of $D_{q,qr}$. Its nearest possible pole has modulus $(qr)^{-1} > q^{-1}$, so choose R strictly between these two radii. The polynomials converge uniformly on $|\zeta| \leq R$. Hence $D_N(v)\Omega_q \rightarrow \eta_{qr}$ in norm. On the output sector,

$$u_- D_N(v)\Omega_q = C_q \delta_{m,-1} D_N(q^{-1}\zeta) \chi_q(q\zeta), \quad (6.7)$$

using $(1 + \zeta)\chi_q(q^{-1}\zeta) = \chi_q(q\zeta)$. This converges in norm to the displayed state. In particular no action is being defined by crossing a pole and choosing a boundary value afterwards. Finally,

$$\frac{1 + \zeta}{1 + r\zeta} = 1 + \sum_{n \geq 1} (1 - r)(-r)^{n-1} \zeta^n \quad (6.8)$$

has squared norm $1 + (1 - r)^2/(1 - r^2) = 2/(1 + r)$. $\square$

6.3 The full norm, including its contact

For $d > 0$, set

$$\beta_{q;r,d}^2 = \frac{dr(1+r)}{(1-r)(1-q^2r^2)}, \quad \Phi_{q;r,d}[h] = \beta_{q;r,d} h(-q^{-1}v) u_- D_{q,qr}(v) \Omega_q \quad (6.9)$$

for a finite Laurent polynomial h . On the output sector $m = -1$, $v = q\zeta$, so $-q^{-1}v = -\zeta$. This normalization of the electric argument is part of the source prescription.

Theorem 6.2 (Schur realization of the positive local prime reference). *With $z = -\zeta$, the preparation (6.9) satisfies*

$$\langle \Phi_{q;r,d}[h], \Phi_{q;r,d}[k] \rangle = \int_{S^1} \overline{h(z)} k(z) w_{r,d}(z) \frac{d\theta}{2\pi}, \quad (6.10)$$

$$w_{r,d}(z) = \frac{dr(1+r)}{1-r} \frac{|1-z|^2}{|1-rz|^2} = \kappa_{r,d} - d \sum_{n \geq 1} r^n (z^n + z^{-n}), \quad (6.11)$$

$$\kappa_{r,d} = \frac{2dr}{1-r}. \quad (6.12)$$

The series converges absolutely and uniformly. In the Laurent basis,

$$\langle \Phi[z^j], \Phi[z^k] \rangle = \begin{cases} \kappa_{r,d}, & j = k, \\ -dr^{|j-k|}, & j \neq k. \end{cases} \quad (6.13)$$

For each prime p , setting $r = p^{-1/2}$ and $d = \log p$ supplies every single-prime off-diagonal coefficient in the Weil form. The same fixed q works for all these preparations.

Proof. Lemma 6.1 and (6.9) give the first expression for $w_{r,d}$ directly from the physical norm. To expand it, use the Poisson-kernel identity

$$\frac{1 - r^2}{|1 - rz|^2} = 1 + \sum_{n \geq 1} r^n (z^n + z^{-n}) \quad (|z| = 1). \quad (6.14)$$

Subtracting this identity from $(1+r)/(1-r)$ and multiplying by d gives (6.11). Integration against $\overline{z^j} z^k$ gives (6.13). Finally $a_p = q/\sqrt{p} < q$, so the graph-domain proof applies to every prime at the same q . All operators needed after the magnetic action are bounded on its fixed output sector for Laurent polynomial inputs. $\square$

The source normalization β uses the assigned arithmetic boundary data. It determines the entire multiplier; no independent contact is subtracted. Since $w_{r,d}$ is bounded, (6.10) extends by density from Laurent polynomials to $L^2(S^1)$ as a bounded source map into the output sector.

Corollary 6.3 (Sharp contact in this stationary scalar class). *A continuous scalar circle multiplier with nonzero Fourier coefficients $-dr^{|n|}$ can be nonnegative only if its constant coefficient is at least $\kappa_{r,d}$. The multiplier in (6.11) attains that value.*

Proof. The multiplier with constant coefficient C equals $C - d \sum_{n \geq 1} r^n (z^n + z^{-n})$. Its value at $z = 1$ is $C - 2dr/(1 - r)$, which must be nonnegative. Equation (6.11) is nonnegative and vanishes at $z = 1$. $\square$

The same statement holds for the corresponding whole-line stationary multiplier. It is not a claim of optimality over all finite-interval or nonstationary realizations. This is the known positive local reference; the result here supplies its Schur operator factor and checked domain.

6.4 An explicit lift to continuous arithmetic inputs

The circle coordinate is not, by itself, the real logarithmic input. To give an actual source map, define

$$(\mathcal{B}_d F)_j(\theta) = d^{-1/2} \widehat{F}((\theta + 2\pi j)/d), \quad 0 \leq \theta < 2\pi, \quad j \in \mathbb{Z}. \quad (6.15)$$

This is a unitary map from $L^2(\mathbb{R})$ to $L^2(S^1, d\theta/(2\pi); \ell^2(\mathbb{Z}))$. Indeed the intervals $[2\pi j/d, 2\pi(j+1)/d]$ partition the Fourier line, and (2.1) gives equality of norms. Conversely an arbitrary square-integrable family specifies a Fourier transform interval by interval, giving surjectivity.

Apply the extended source of Theorem 6.2 to each component of $\mathcal{B}_d F$ in an orthogonal copy of $\mathcal{H}$. The resulting positive norm is

$$\begin{aligned} B_{r,d}[F] &= \frac{1}{2\pi} \int_{\mathbb{R}} w_{r,d}(e^{id\tau}) |\widehat{F}(\tau)|^2 d\tau \\ &= \kappa_{r,d} \|F\|^2 - 2d \sum_{n \geq 1} r^n \operatorname{Re} \langle F, U_{nd} F \rangle. \end{aligned} \quad (6.16)$$

The Fourier sign for U_d does not change the expression, because both orientations occur. For $F = E_L f$, only $nd < L$ remains.

This construction adds a specified multiplicity space and preparation. That multiplicity is not identified with the magnetic-charge label already present in (6.1). No particular four-dimensional interface has yet been shown to implement (6.15). The result is an exact positive Hilbert-space realization containing an elementary Schur operator factor. A full field-theoretic interpretation of the lift is part of the open matching problem.

6.5 Which output states the insertion can reach

The preparation of Lemma 6.1 produces one particular state on the output sector. It is worth recording how much of that state is forced by the representation and how much is a choice of dressing, because the answer governs what a weight identity in this sector can be taken to show.

On the zero-charge sector an electric preparation $\eta = D(v)\Omega_q$ has wavefunction $\eta(w) = C_q D(w)\chi_q(w)$, and (6.2) gives

$$(u_- \eta)_{-1}(\zeta) = (1 + \zeta) \eta(q^{-1}\zeta). \quad (6.17)$$

Proposition 6.4 (Reachable output states). *Let ψ be analytic on $\{|\zeta| < \rho\}$ for some $\rho > 1$, and set $\eta(w) = \psi(qw)/(1 + qw)$. Then $\eta = D(v)\Omega_q$ for*

$$D(w) = \frac{\psi(qw)(-q^3w; q^2)_\infty}{C_q}, \quad (6.18)$$

which is analytic on $\{|w| < \rho/q\}$ with $\rho/q > q^{-1}$; the graph-domain argument of Lemma 6.1 applies to D unchanged, and the resulting output state on the sector $m = -1$ is ψ . Conversely every output state arises in this way. Hence the states reachable by one negative magnetic insertion, from an electric preparation whose dressing is analytic on a disk of radius exceeding q^{-1} , are exactly the functions analytic on some disk of radius exceeding one.

Proof. Since $\chi_q(w)^{-1} = (-qw; q^2)_\infty = \prod_{j \geq 0} (1 + q^{2j+1}w)$ has $(1 + qw)$ as its $j = 0$ factor,

$$C_q D(w) \chi_q(w) = \psi(qw) \frac{\prod_{j \geq 1} (1 + q^{2j+1}w)}{\prod_{j \geq 0} (1 + q^{2j+1}w)} = \frac{\psi(qw)}{1 + qw} = \eta(w).$$

The tail product is entire, so the singularities of D are those of $\psi(qw)$, at modulus at least ρ/q . Choose R strictly between q^{-1} and ρ/q . The Taylor polynomials D_N converge uniformly on $|w| \leq R$, hence $D_N(q^{-1}\zeta) \rightarrow D(q^{-1}\zeta)$ uniformly on $|\zeta| \leq qR$, a disk containing the closed unit disk, and $\chi_q(q\zeta)$ is bounded there because its nearest pole has modulus q^{-2} ; the proof of Lemma 6.1 then applies verbatim, with $D_N(v)\Omega_q \rightarrow \eta$ in norm and $u_- D_N(v)\Omega_q \rightarrow \delta_{m,-1}\psi$ in norm. Finally (6.17) gives $(1 + \zeta)\eta(q^{-1}\zeta) = (1 + \zeta)\psi(\zeta)/(1 + \zeta) = \psi(\zeta)$, and the same equation determines η from ψ . $\square$

The factor $(1 + \zeta)$ inserted by u_- is therefore not a constraint on the output. It is cancelled, for every target ψ , by the $j = 0$ factor of the vacuum's Pochhammer, which the dressing (6.3) removes and another dressing may retain. The $(1 + \zeta)$ surviving in (6.6) is the $j = 0$ factor of that dressing's own numerator $(-qw; q^2)_\infty$ evaluated at $w = q^{-1}\zeta$, not a feature of the operator. No pole is crossed in either case: the cancellation happens between the dressing's analyticity and the identity $(1 + \zeta)\chi_q(q^{-1}\zeta) = \chi_q(q\zeta)$ already used in (6.7).

Two consequences. The mirror reference of Section 7.5 is reachable, which is Corollary 7.10 below. And producing a prescribed weight in this sector is not by itself evidence of arithmetic structure in the representation, since any weight whose square root is analytic past the unit circle can be produced by choosing the dressing. What Theorem 6.2 establishes beyond the weight is that one fixed q serves every prime and that the insertion is justified on a checked domain. Candidate preparations should be assessed on those points, and on the compression of Section 7.6, rather than on the appearance of a prime weight in a positive sector.

7 Magnetic domains and joint-prime gluing

7.1 What higher magnetic powers would require

The domain statement for one insertion is sufficient for Theorem 6.2. It does not place the prepared state in the domain of every negative magnetic power.

Proposition 7.1 (Formal higher-power coefficient criterion). *For $k \geq 1$, iteration of the coefficient action gives the formal series*

$$(u_-^k \eta_a)_{-k}(\zeta) = \sqrt{1 - a^2} \frac{\prod_{j=0}^{k-1} (1 + q^{k-1-2j}\zeta)}{1 + aq^{-k}\zeta}. \quad (7.1)$$

Its Taylor coefficient sequence belongs to ℓ^2 if and only if

$$a < q^k \quad \text{or} \quad a \in \{q, q^3, \dots, q^{2k-1}\}. \quad (7.2)$$

For fixed $0 < a < 1$, square summability of all these formal outputs is equivalent to $a = q^{2j+1}$ for some integer $j \geq 0$.

Proof. Induction in (6.2) gives (7.1). For every coefficient of degree $n \geq k$, expanding the denominator shows that the coefficient equals

$$\sqrt{1-a^2}(-aq^{-k})^n \prod_{j=0}^{k-1} (1 - q^{2j+1}/a). \quad (7.3)$$

If the product is nonzero, the geometric tail is square summable exactly when $aq^{-k} < 1$. If the product vanishes, the numerator cancels the denominator and the output is a polynomial. This proves (7.2). For all k , the inequality eventually fails at any fixed $a > 0$, so an exceptional value is necessary. If $a = q^{2j+1}$, it satisfies the strict inequality for $k \leq j$ and the cancellation condition for $k \geq j+1$, proving sufficiency for the formal coefficient statement. $\square$

The exceptional square-summable outputs do not by themselves prove membership in powers of a chosen closed operator extension. A divergent coefficient tail is, however, a necessary domain obstruction whenever the prescribed action on Fourier coefficients is retained. This distinction prevents meromorphic continuation across a pole from being mistaken for an operator-domain argument.

For the prime states $a_p = qp^{-1/2}$, requiring all negative powers would force $p^{-1/2} = q^{2j_p}$. This cannot hold simultaneously for $p = 2$ and $p = 3$, since it would imply a nontrivial equality between an integer power of 2 and an integer power of 3. It does not affect the single insertion already proved. More generally, for any fixed finite $K > 1$, choosing $q^{K-1} > 2^{-1/2}$ puts every prime state in the strictly decaying regime for all powers through K . In that regime the Taylor-approximation argument of Lemma 6.1 extends through the finite sequence of shifts. A proposed junction must specify the finite words or actual operator domains it uses; a field theory need not put every state in the domain of every unbounded power.

7.2 A two-prime obstruction for fixed output vectors

For each prime write $r_p = p^{-1/2}$, $d_p = \log p$, and

$$A_p(z) = \frac{1-z}{1-r_p z} = 1 - (1-r_p) \sum_{n \geq 1} r_p^{n-1} z^n, \quad c_p = \frac{d_p r_p (1+r_p)}{1-r_p}. \quad (7.4)$$

The norm of $\sqrt{c_p} A_p$ is the multiplier (6.11). Consider a proposed coherent sum with fixed unit output vectors v_2, v_3 :

$$F \mapsto \hat{F}(\tau) \left[\sqrt{c_2} A_2(e^{id_2 \tau}) v_2 + \sqrt{c_3} A_3(e^{id_3 \tau}) v_3 \right], \quad (7.5)$$

with output measure $d\tau/(2\pi)$.

Proposition 7.2 (Mixed atom from constant-vector gluing). *In (7.5), the coefficient of $e^{i(\log 3 - \log 2)\tau}$ in the full norm multiplier is*

$$\sqrt{c_2 c_3} (1-r_2)(1-r_3) \langle v_2, v_3 \rangle. \quad (7.6)$$

Consequently this source, together with independently added gamma, contact and pole terms, cannot equal the two-prime Weil form on $I_{5/4}$ unless $v_2 \perp v_3$. In the orthogonal case its prime part is exactly the sum of the separate positive references.

Proof. The first cross term in the squared norm is $\sqrt{c_2 c_3} A_2(e^{id_2 \tau}) A_3(e^{id_3 \tau}) \langle v_2, v_3 \rangle$. Its first non-constant coefficient from each factor gives (7.6). A second pair of nonnegative indices with $n \log 3 - m \log 2 = \log 3 - \log 2$ would give $3^{n-1} = 2^{m-1}$ and hence $n = m = 1$. The reverse cross term cannot give this positive frequency with nonnegative indices, and the individual squared norms contain only integer multiples of d_2 or d_3 .

The Fourier coefficient series of each factor is absolutely summable; the cross kernel is therefore a finite-variation atomic measure. Other mixed lengths may accumulate, but its atom at $\log(3/2)$ has the specified nonzero mass if the overlap is nonzero. The Weil target has no prime-power atom there. Its gamma and pole kernels are smooth near this nonzero separation, and its contact is supported at zero, so none can cancel the atom. Since $0 < \log(3/2) < 5/4$, restriction to $I_{5/4}$ still detects it. Equality of all polarized test-function pairings would imply equality of their distribution kernels, a contradiction. A purely imaginary overlap gives an imaginary antisymmetric kernel; complex tests still detect it. Thus the overlap must vanish. $\square$

The proposition does not apply to arbitrary coherent gluing. A joint operator-valued junction may have additional terms which cancel mixed returns as an identity. Producing and evaluating such a junction is a new source construction. The failed fixed-vector ansatz gives no reason to estimate its unwanted atoms as a small residual.

7.3 The full comparison, with no omitted contact or pole

Let $\mathcal{P}$ be any finite set containing all primes $p < e^L$. Equations (2.9) and (6.16) give the exact identity

$$\begin{aligned} Q_L[f] &= K[E_L f] + \sum_{p \in \mathcal{P}} B_{r_p, d_p}[E_L f] \\ &\quad + \left(w_0 - \sum_{p \in \mathcal{P}} \kappa_{r_p, d_p} \right) \|f\|^2 + P_L[f]. \end{aligned} \quad (7.7)$$

Each term in the first line is independently positive. The second line has not been realized as part of their joint norm. Extra primes in $\mathcal{P}$ contribute only contacts on this interval, which cancel algebraically against the displayed subtraction. This does not make the subtraction an independently positive operation.

Thus the dressed result arrives at the same contact and pole problem as the conservative prime reference. It supplies a concrete Schur preparation and removes the earlier undressed-state mismatch, but establishes no larger positivity interval. Unitary repackaging of the orthogonal gamma and prime outputs leaves (7.7) unchanged. The next calculation must determine contact, poles and prime interference within one complete positive preparation.

7.4 Interference is necessary, and the two-channel architecture fails

Equation (7.7) presents Q_L as a sum of two independently positive forms and a residual. Write

$$A_L[f] = K[E_L f], \quad B_L[f] = \sum_{p \in \mathcal{P}} B_{r_p, d_p}[E_L f], \quad R_L[f] = \left(w_0 - \sum_{p \in \mathcal{P}} \kappa_{r_p, d_p} \right) \|f\|^2 + P_L[f], \quad (7.8)$$

so that $Q_L = A_L + B_L + R_L$. The residual is not a positive form, and the first observation is that it cannot be repaired by adding more positive channels.

Proposition 7.3 (No orthogonal augmentation). *Let $c < 0$. On the subspace $\{f \in C_c^\infty(I_L) : \int_{I_L} f \cosh(x/2) = \int_{I_L} f \sinh(x/2) = 0\}$, which has codimension two, the form $c\|f\|^2 + P_L[f]$ equals $c\|f\|^2 < 0$. Hence R_L is not the Gram form of any further channel adjoined orthogonally to a preparation whose norm is $A_L + B_L$.*

Proof. P_L vanishes on that subspace by (2.8), and the subspace is the common kernel of two linear functionals on an infinite-dimensional space. $\square$

The remaining possibility is interference: the archimedean and arithmetic parts of a preparation must fail to be orthogonal. The next elementary bound says how much interference is available.

Lemma 7.4 (Interference bound). *Let G, S be complex-linear maps from a complex vector space V into a Hilbert space $\mathcal{H}$. Put $A[f] = \|Gf\|^2$, $B[f] = \|Sf\|^2$ and $C[f] = \|(G + S)f\|^2 - A[f] - B[f]$. Then*

$$tA[f] + t^{-1}B[f] + C[f] \geq 0 \quad (t > 0, f \in V), \quad (7.9)$$

equivalently $-C[f] \leq 2\sqrt{A[f]B[f]}$, that is $\|(G + S)f\|^2 \geq (\sqrt{A[f]} - \sqrt{B[f]})^2$.

Proof. $C[f] = 2\operatorname{Re}\langle Gf, Sf \rangle$, so $tA[f] + t^{-1}B[f] + C[f] = \|\sqrt{t}Gf + t^{-1/2}Sf\|^2 \geq 0$. Minimizing the left side over $t > 0$ at $t = \sqrt{B/A}$ gives the other two forms, and replacing S by $-S$ leaves the statement unchanged. $\square$

At $t = 1$ the bound (7.9) is only the positivity of Q_L . Other values of t are strictly stronger, and they exclude the architecture of (7.7).

Theorem 7.5 (Exclusion of the archimedean/prime two-channel source). *Let $L = 1$, or $L = 5/4$, or let L be sufficiently large. There are no complex-linear maps G, S from $C_c^\infty(I_L)$ into a common positive Hilbert space with*

$$\|Gf\|^2 = A_L[f], \quad \|Sf\|^2 = B_L[f], \quad \|(G \pm S)f\|^2 = Q_L[f] \quad (7.10)$$

for every $f \in C_c^\infty(I_L)$.

Proof. For such a pair the interference form is $C = Q_L - A_L - B_L = R_L$, so Lemma 7.4 requires

$$N_L(t)[f] := tA_L[f] + t^{-1}B_L[f] + R_L[f] \geq 0 \quad (7.11)$$

for every $t > 0$. Each of B_L , P_L and $\|\cdot\|^2$ is bounded and A_L is the closed gamma energy, so $N_L(t)$ is continuous for the energy norm of $\mathcal{D}_{\gamma,L}$. Moreover $C_c^\infty(I_L)$ is dense in $\mathcal{D}_{\gamma,L}$: if g is bounded by one and supported in a set of measure 2ε then $2\|g\|^2 - 2\operatorname{Re}\langle g, U_r g \rangle \leq 2\min(r, \varepsilon)$ pointwise in r , so $K[g] = O(\varepsilon \log(1/\varepsilon))$, and cutting $\mathbf{1}_{I_L}$ off smoothly within ε of each endpoint changes the energy by that much. It therefore suffices to exhibit one $f \in \mathcal{D}_{\gamma,L}$ and one $t > 0$ with $N_L(t)[f] < 0$.

Take $f = \mathbf{1}_{I_L}$, so that $\|f\|^2 = L$ and $\langle F, U_r F \rangle = (L - r)^+$. Then

$$A_L[\mathbf{1}_{I_L}] = \int_0^\infty [2L - 2(L - r)^+] n_\gamma(r) dr = 2 \sum_{n \geq 0} \frac{1 - e^{-a_n L}}{a_n^2} \leq \frac{1}{2} \psi'(\frac{1}{4}), \quad (7.12)$$

using $\int_0^L r e^{-ar} dr + L e^{-aL}/a = (1 - e^{-aL})/a^2$ and $\sum_{n \geq 0} a_n^{-2} = \frac{1}{4} \psi'(\frac{1}{4})$. Likewise

$$B_L[\mathbf{1}_{I_L}] = \sum_p \left[\kappa_{r_p, d_p} L - 2d_p \sum_{nd_p < L} r_p^n (L - nd_p) \right], \quad P_L[\mathbf{1}_{I_L}] = 32 \sinh^2(L/4), \quad (7.13)$$

the second because $\int_{I_L} \sinh(x/2) dx = 0$ and $\int_{I_L} \cosh(x/2) dx = 4 \sinh(L/4)$.

For $L = 5/4$ the active translations are $\log 2$ and $\log 3$, and (7.12)–(7.13) give

$$A_L = 4.302142, \quad B_L = 7.197439, \quad P_L = 3.228059, \quad \kappa_{r_2, d_2} + \kappa_{r_3, d_3} = 6.348275,$$

so $R_L = -11.422514$ and $N_L(1) = Q_L[\mathbf{1}_{I_L}] = 0.077067 > 0$, while at $t = 13/10$

$$N_L(13/10) = 5.592785 + 5.536491 - 11.422514 = -0.293238 < 0.$$

For $L = 1$ the same computation with the eight-cell step function $f = (1, 1, 0, -1, -1, 0, 1, 1)$ on cells of width $1/8$ and $t = 3/4$ gives $N_L(t)[f] = -0.106621 < 0$, again against $Q_L[f] = 0.187826 > 0$. Both evaluations are finite closed forms in $\psi'(1/4)$, $\sinh$, $\log 2$ and $\log 3$; they are carried out with an explicit tail bound for the gamma series in Appendix D.

For large L take $f = \mathbf{1}_{I_L}$ again. By (7.12) $A_L \leq \frac{1}{2}\psi'(\frac{1}{4})$ uniformly in L , and by (7.13) $B_L \leq L \sum_{p < e^L} \kappa_{r_p, d_p}$, since the subtracted correlations are nonnegative. The prime number theorem and partial summation give $\sum_{p < x} \kappa_{r_p, d_p} = 4\sqrt{x}(1 + o(1))$, so

$$2\sqrt{A_L B_L} = O(L^{1/2} e^{L/4}), \quad -R_L = \left(\sum_{p < e^L} \kappa_{r_p, d_p} - w_0 \right) L - 32 \sinh^2(L/4) = 4(L-2)e^{L/2}(1 + o(1)).$$

Hence $-R_L > 2\sqrt{A_L B_L}$ for all large L , and (7.9) fails by a factor tending to infinity. $\square$

Remark 7.6. The failure is not confined to the two certified intervals. Evaluating (7.11) at $f = \mathbf{1}_{I_L}$ and $t = \sqrt{B_L/A_L}$ on a grid of step $1/10$ shows $N_L(t) < 0$ at every $L \geq 11/10$ up to $L = 13/2$; at $L = 1$ the indicator alone gives $+0.054$ and the eight-cell step function above is needed. These evaluations are recorded in Appendix D.

Theorem 7.5 excludes an architecture, not the identities it is assembled from. Theorem 6.2 remains correct; what fails is the plan of adjoining to it a separately positive archimedean channel and recovering (2.9). The exclusion is specific to the pair (7.8): a preparation whose channel norms are different forms A', B' with $A' + B' + C' = Q_L$ is not covered, and (7.9) is then the first test such a preparation must pass. A source in which archimedean and prime data are not separated into two channels at all is untouched by the argument. Proposition 7.2 and Theorem 7.5 are complementary. The first forbids a coherent sum with fixed output vectors, because of the mixed translations it creates. The second forbids the orthogonal-channel bookkeeping of (7.7) however the two channels are subsequently correlated.

7.5 The mirror reference and the subtraction form

Lemma 7.4 bounds the norm of a *sum* from below. It says nothing about a difference $\|Gf\|^2 - \|\Pi Gf\|^2$, and Proposition 7.3 says that a difference is what is needed. This subsection shows that the target has such a presentation, with the arithmetic confined to the subtracted term.

The positive reference of Theorem 6.2 has the least contact compatible with prime atoms of one sign (Corollary 6.3). There is a second elementary weight with the same atoms and the opposite sign.

Proposition 7.7 (Mirror prime reference). *For $0 < r < 1$ and $d > 0$ set*

$$\tilde{w}_{r,d}(z) = \frac{dr(1-r)}{1+r} \frac{|1+z|^2}{|1-rz|^2}, \quad |z| = 1. \quad (7.14)$$

Then $\tilde{w}_{r,d} \geq 0$ and

$$\tilde{w}_{r,d}(z) = \tilde{\kappa}_{r,d} + d \sum_{n \geq 1} r^n (z^n + z^{-n}), \quad \tilde{\kappa}_{r,d} = \frac{2dr}{1+r}. \quad (7.15)$$

The constant $\tilde{\kappa}_{r,d}$ is the least one for which a continuous scalar multiplier with Fourier coefficients $+dr^{|n|}$ is nonnegative, and

$$w_{r,d} + \tilde{w}_{r,d} = \frac{4dr}{1-r^2}, \quad \tilde{\kappa}_{r,d} < \kappa_{r,d} = \frac{2dr}{1-r}. \quad (7.16)$$

Proof. Nonnegativity is clear from (7.14). Write $c = z + z^{-1}$ and use the Poisson identity (6.14). The constant Laurent coefficient of $(2+c)/|1-rz|^2$ is $(2+2r)/(1-r^2) = 2/(1-r)$, and for $m \geq 1$ its z^m coefficient is

$$\frac{2r^m + r^{m-1} + r^{m+1}}{1-r^2} = \frac{r^{m-1}(1+r)^2}{1-r^2} = \frac{r^{m-1}(1+r)}{1-r}.$$

Multiplying by $dr(1-r)/(1+r)$ gives (7.15). Adding (6.11) cancels every nonconstant coefficient and leaves $\kappa_{r,d} + \tilde{\kappa}_{r,d} = 4dr/(1-r^2)$. For minimality, a multiplier $C + d \sum_{n \geq 1} r^n (z^n + z^{-n})$ takes at $z = -1$ the value $C - 2dr/(1+r)$, which must be nonnegative; (7.14) vanishes there. $\square$

Applying the lift (6.15) to (7.15) exactly as in Section 6 gives a positive form on $C_c^\infty(I_L)$,

$$\tilde{B}_{r,d}[F] = \tilde{\kappa}_{r,d} \|F\|^2 + 2d \sum_{n \geq 1} r^n \operatorname{Re} \langle F, U_{nd} F \rangle \geq 0. \quad (7.17)$$

Theorem 7.8 (Prime-free subtraction form of the target). *Let $\mathcal{P}$ be any finite set of primes containing every $p < e^L$. Then*

$$Q_L[f] = A_L[f] - \sum_{p \in \mathcal{P}} \tilde{B}_{r_p, d_p}[E_L f], \quad (7.18)$$

$$A_L[f] = K[E_L f] + \left(w_0 + \sum_{p \in \mathcal{P}} \tilde{\kappa}_{r_p, d_p} \right) \|f\|^2 + P_L[f]. \quad (7.19)$$

The form A_L involves no prime translation: away from the diagonal its kernel is $-n_\gamma(|x-y|) + 2 \cosh((x-y)/2)$, which is smooth, and its only arithmetic input is the constant in (7.19). Primes outside e^L contribute only contacts, which cancel against that constant.

Proof. By (7.17) the prime term of (2.9) is $-\sum_p (\tilde{B}_{r_p, d_p} - \tilde{\kappa}_{r_p, d_p} \|f\|^2)$, since for $F = E_L f$ the correlations with $nd_p \geq L$ vanish and the coefficient of $\operatorname{Re} \langle F, U_{nd_p} F \rangle$ is $-2d_p r_p^n$ in both. Substituting in (2.9) gives (7.18). The kernel statement is (2.4) and (2.8). $\square$

Corollary 7.9 (The criterion as a domination, and as a compression). *For each $L > 0$, $Q_L \geq 0$ if and only if $\sum_p \tilde{B}_{r_p, d_p} \preceq A_L$ on $C_c^\infty(I_L)$. If in addition $A_L[f] = \|Gf\|^2$ for a complex-linear G into a positive Hilbert space and $\sum_p \tilde{B}_{r_p, d_p}[E_L f] = \langle Gf, \Pi Gf \rangle$ for a positive contraction Π on the closure of the range of G , then*

$$Q_L[f] = \|(1-P)V Gf\|^2 \quad (7.20)$$

for an isometry V and an orthogonal projection P on a larger space. In particular Q_L is realized as a compression, and Lemma 7.4 places no obstruction on this presentation.

Proof. The equivalence is (7.18). Given Π , we have $Q_L[f] = \langle Gf, (1 - \Pi)Gf \rangle$, and $1 - \Pi$ is a positive contraction; by the dilation theorem $1 - \Pi = V^*(1 - P)V$ for an isometry V and a projection P , which gives (7.20). Lemma 7.4 concerns $\|(G + S)f\|^2$ and does not apply to (7.20). $\square$

Two features make (7.18) a better construction target than (7.7), and one numerical observation says what is still hard.

The mirror contact is much smaller. Since $\tilde{\kappa}_{r,d}/\kappa_{r,d} = (1 - r)/(1 + r)$, the saving is concentrated on the small primes, which is exactly where the budget of (7.7) was lost:

L	$\sum_p \kappa_{r_p, d_p}$	$\sum_p \tilde{\kappa}_{r_p, d_p}$	$w_0 + \sum_p \tilde{\kappa}_{r_p, d_p}$
1	3.3468	0.5742	-4.7980
5/4	6.3483	1.3785	-3.9937
2	11.3172	3.4406	-1.9315
3	18.9237	7.8704	+2.4983
4	30.3436	16.0103	+10.6381

The constant in (7.19) changes sign near $L = 2.6$; beyond that point the prime-free form has a positive contact.

The prime-free form is not marginal. In a Fourier–Galerkin space of 61 modes on I_L , the least eigenvalue of A_L relative to $\|\cdot\|^2$ is 0.258 at $L = 1$, 0.799 at $L = 5/4$, 0.556 at $L = 3/2$, 2.13 at $L = 2$ and 22.4 at $L = 5$, while the same computation gives $\lambda_{\min}(Q_L) \leq 10^{-6}$ for $L \geq 1$. The difficulty has moved out of the source and into the subtraction.

The domination is saturated. Writing $\Pi_L = A_L^{-1/2}(\sum_p \tilde{B}_{r_p, d_p})A_L^{-1/2}$, the criterion of Corollary 7.9 is $\|\Pi_L\| \leq 1$, and the same computation returns $\|\Pi_L\| = 1$ to within 10^{-7} with the number of eigenvalues above 0.9 growing from two at $L = 4/5$ to thirteen at $L = 2$. That is expected: $\|\Pi_L\| = 1 - \min \text{spec}(Q_L; A_L)$ and Q_L is nearly degenerate. For contrast, the corresponding quotient for (7.7), which asks $\sum_p B_{r_p, d_p} \preceq K$, equals 1.75 at $L = 5/4$, 3.57 at $L = 2$ and 26.1 at $L = 5$. A subspace value above one certifies that a domination fails, so those numbers are the quantitative form of Theorem 7.5; a subspace value below one certifies nothing, so the first row is evidence and not a proof.

Corollary 7.10 (Schur realization of the mirror reference). *Fix $0 < q < 1$ and $0 < r < 1$ and put $a = qr$. The dressing*

$$\tilde{D}_{q,a}(w) = \frac{\sqrt{1 - a^2} (1 - qw) (-q^3 w; q^2)_\infty}{C_q (1 + aw)}, \quad (7.21)$$

which is (6.3) with the sign of its first Pochhammer factor reversed, is analytic except for a simple pole at $w = -a^{-1}$, of modulus $(qr)^{-1} > q^{-1}$. Its electric state is $\tilde{\eta}_{qr}(w) = \sqrt{1 - a^2}(1 - qw)/((1 + qw)(1 + aw))$, one negative magnetic insertion gives

$$(u - \tilde{\eta}_{qr})_{-1}(\zeta) = \sqrt{1 - q^2 r^2} \frac{1 - \zeta}{1 + r\zeta}, \quad (7.22)$$

and with $\tilde{\beta}_{q;r,d}^2 = dr(1 - r)/((1 + r)(1 - q^2 r^2))$, which is (6.9) with $(1 + r)/(1 - r)$ inverted, the preparation $\tilde{\Phi}_{q;r,d}[h] = \tilde{\beta}_{q;r,d} h(-q^{-1}v) u - \tilde{D}_{q,a}(v) \Omega_q$ has pairing $\int_{S^1} \bar{h}(z) k(z) \tilde{w}_{r,d}(z) d\theta/2\pi$ with $\tilde{w}_{r,d}$ as in (7.14). As in Theorem 6.2, one fixed q serves every prime.

Proof. Apply Proposition 6.4 with $\psi(\zeta) = (1 - \zeta)/(1 + r\zeta)$, which is analytic on $|\zeta| < r^{-1}$ and $r^{-1} > 1$. Then (6.18) gives $D(w) \propto (1 - qw)(-q^3 w; q^2)_\infty/(1 + qrw)$, which is (7.21), and the output state is ψ up to the stated normalization. With $z = -\zeta$ one has $|1 - \zeta|^2 = |1 + z|^2$ and $|1 + r\zeta|^2 = |1 - rz|^2$, and $\tilde{\beta}_{q;r,d}^2(1 - q^2 r^2) = dr(1 - r)/(1 + r)$ is the constant in (7.14). $\square$

Both references are therefore states of the same representation at the same fixed q , and the mirror does not require its own protected sector. The two output states $\psi_{\pm} = (1 \pm \zeta)/(1 + r\zeta)$ are orthogonal, with $\|\psi_{\pm}\|^2 = 2/(1 \pm r)$, and in the basis $(1 + r\zeta)^{-1}$, $\zeta(1 + r\zeta)^{-1}$ their Gram matrix is $(1 - r^2)^{-1} \begin{pmatrix} 1 & -r \\ -r & 1 \end{pmatrix}$, positive definite. By Proposition 6.4 the freedom exhibited here is freedom of weights alone; a positive sector supplies no null vectors, and Section 8.2 says why none are needed.

7.6 The target as a compression of the gamma energy

Corollary 7.9 presents Q_L as a compression of a source for A_L . That presentation requires two things not in hand: a source for A_L , and the inequality $A_L \geq 0$, which is (8.1) and is settled only outside a compact window (Section 8.2). Splitting the pole form by parity removes both requirements.

Proposition 7.11 (Compression of the gamma energy). *Let $\mathcal{P}$ be any finite set of primes containing every $p < e^L$, put $c_L = w_0 + \sum_{p \in \mathcal{P}} \tilde{\kappa}_{r_p, d_p}$, $c_+ = \max(c_L, 0)$ and $c_- = \max(-c_L, 0)$, and define on $C_c^\infty(I_L)$*

$$K_+[f] = K[E_L f] + c_+ \|f\|^2 + 2|\langle \cosh(x/2), f \rangle|^2, \quad (7.23)$$

$$T_L[f] = c_- \|f\|^2 + 2|\langle \sinh(x/2), f \rangle|^2 + \sum_{p \in \mathcal{P}} \tilde{B}_{r_p, d_p}[E_L f]. \quad (7.24)$$

Then $Q_L = K_+ - T_L$. The form K_+ is a norm, $K_+[f] = \|Gf\|^2$ for the source

$$Gf = \left(b(D)^{1/2} E_L f, c_+^{1/2} f, \sqrt{2} \langle \cosh(x/2), f \rangle \right) \in L^2(\mathbb{R}) \oplus L^2(I_L) \oplus \mathbb{C}, \quad (7.25)$$

where $b(D)^{1/2}$ is the Fourier multiplier with symbol $b(\tau^2)^{1/2}$; and $T_L \geq 0$, each of its three terms separately. Consequently $Q_L \geq 0$ on $C_c^\infty(I_L)$ if and only if $T_L \preceq K_+$, and in that case

$$Q_L[f] = \langle Gf, (1 - \Pi)Gf \rangle \quad (7.26)$$

for the positive contraction Π on the closure of the range of G determined by $\langle Gf, \Pi Gg \rangle = T_L(f, g)$.

Proof. Substituting (2.17) for P_L in (7.19), and moving the constant to the side fixed by its sign, turns (7.18) into $Q_L = K_+ - T_L$; nothing is cancelled or estimated. K_+ is a norm because $b \geq 0$ in (2.6), so $K[F] = \|b(D)^{1/2} F\|^2$ by (2.5), while the other two terms are a nonnegative multiple of $\|f\|^2$ and a rank-one square. $T_L \geq 0$ because $c_- \geq 0$, the second term is a rank-one square, and $\tilde{B}_{r_p, d_p} \geq 0$ by (7.17). For the last statement, $t(Gf, Gg) = T_L(f, g)$ is well defined on the range of G precisely because $T_L \preceq K_+$: if $Gf = Gf'$ then $K_+[f - f'] = 0$, hence $T_L[f - f'] = 0$. It satisfies $0 \leq t(u, u) \leq \|u\|^2$ there, so the Riesz representation supplies a positive contraction Π with $t(u, v) = \langle u, \Pi v \rangle$, and (7.26) follows. $\square$

Three differences from Corollary 7.9 are worth stating.

The dominating side has an explicit source. (7.25) is a Fourier multiplier together with two elementary coordinates, and its positivity is that of the digamma partial-fraction expansion (2.6), which is prior to any arithmetic input. That is what distinguishes it from the observation in Section 3 that $\Phi = Q_L^{1/2} f$ satisfies Definition 3.1 on one interval as soon as $Q_L \geq 0$: there the positivity assumed is the statement to be proved, here it is not. Appendix A realizes the same gamma form as a minimized positive field energy, and a candidate model is being asked to replace (7.25) by a physically defined preparation rather than to supply positivity.

No unproved inequality enters the presentation. Corollary 7.9 needs $A_L \geq 0$ in order to write $A_L[f] = \|Gf\|^2$ at all. Both sides of Proposition 7.11 are positive term by term, at every L , so (8.1)

is not a prerequisite for it. That inequality remains an open question about A_L ; no part of the construction problem now waits on it.

The price is that the subtracted term is no longer purely arithmetic. It carries the contact and the odd half of the pole form as well as every prime. The trade is a dominating form that can be written down, against a subtracted form that mixes archimedean and arithmetic data.

The constant is a step function of L , jumping by $\tilde{\kappa}_{r_p, d_p}$ at $L = \log p$ from $w_0 = -5.37218\dots$, and its partial sums first become positive at $p = 13$:

p	2	3	5	7	11	13	17
c_L after p	-4.7980	-3.9937	-2.9990	-1.9315	-0.8205	+0.2933	+1.3994

So $c_L \leq 0$ exactly for $L < \log 13 = 2.5649\dots$, which is the precise form of the sign change recorded after (7.19); above it the constant joins the source and only helps.

The presentation does not make the problem smaller, and says so quantitatively. Since $\|\Pi\| = 1 - \min \text{spec}(Q_L; K_+)$ and Q_L is nearly degenerate, the domination is saturated. The check program registered against this subsection works in a basis of 40 indicator cells on I_L and gives $\min \text{spec}(Q_L; K_+) = 5.5 \times 10^{-4}$, 4.1×10^{-4} , 3.7×10^{-4} and 3.0×10^{-4} at $L = 1, 3/2, 2$ and 3 .

A step basis is poorly adapted to this quantity, because the transforms of its elements decay only like τ^{-1} and so cannot avoid the lowest zero ordinate. Every Rayleigh quotient is an upper bound for the true value, and a smooth basis does far better. Taking $\phi_k(x) = (1 - (2x/L)^2)^k$ and assembling Q_L from its spectral form (2.14), so that the cancellation recorded in Section 2.4 never occurs, twelve such modes give

$$\min \text{spec}(Q_L; K_+) \leq 6.4 \times 10^{-10}, \quad 2.5 \times 10^{-14}, \quad 1.7 \times 10^{-17}, \quad 1.2 \times 10^{-24} \quad (7.27)$$

at the same four values of L , and the bound continues to fall as the basis is enlarged. The values in (7.27) are recorded in the investigation's notes and are not among the registered check programs; they use arbitrary-precision arithmetic and a truncated sum over zero ordinates, and truncating that sum only lowers them. In the cell basis, with 400 cells, the number of generalized eigenvalues below 10^{-3} is 2, 6, 10 and 14 at $L = 1, 3/2, 2$ and 3 .

A candidate mechanism must therefore produce a *critical* contraction rather than a strict one, whose spectrum piles up at one on a subspace growing with L ; a mechanism yielding a uniform strict contraction, or more generally any hypothesis supplying a margin uniform in L , would be producing a false statement. All of these are one-sided subspace values, as after Corollary 7.9: above one they would certify that the domination fails, below one they certify nothing.

Remark 7.12 (The lowest eigenvalue at small L). Suzuki studies the same lowest eigenvalue directly, and unconditionally [1]. Writing the localized form as $\langle A_a f, f \rangle$ for a self-adjoint A_a on $L^2(-a, a)$ with $a = L/2$, he proves that A_a is the Friedrichs extension of an explicit symmetric operator, that its lowest eigenvalue λ_a is continuous in a , and that for sufficiently small a it is positive, simple, and

$$\lambda_a = \log(1/a) + \mu_1 - \log(2\pi) + \psi(2) - 1 + O(a). \quad (7.28)$$

None of this assumes RH. Its leading behaviour is the archimedean symbol of the present paper evaluated at the interval's own frequency scale: adding (2.7) to (2.10) gives

$$b(\tau^2) + w_0 = \log \frac{|\tau|}{2\pi} + O(|\tau|^{-2}), \quad (7.29)$$

and $\tau \sim 1/a$ reproduces the first two terms of (7.28), the constant included. The positivity in (7.28) is a small-interval statement and does not reach the range in which the domination above is saturated.

7.7 The target as a jump form, and a necessary condition

The archimedean and prime parts of (2.9) are the continuous and atomic parts of one nonnegative measure. Adding and subtracting the value of the prime symbol at $\tau = 0$ puts everything but the pole term into Lévy–Khinchine form.

Proposition 7.13 (Jump form of the target). *For $f \in C_c^\infty(I_L)$ and $F = E_L f$ set*

$$\mu_L = 2n_\gamma(r) dr + 2 \sum_{m \log p < L} (\log p) p^{-m/2} \delta_{m \log p}, \quad \gamma_L = w_0 - 2 \sum_{m \log p < L} (\log p) p^{-m/2}, \quad (7.30)$$

and

$$\mathcal{E}_L[F] = \int_0^\infty [\|F\|^2 - \operatorname{Re}\langle F, U_r F \rangle] d\mu_L(r). \quad (7.31)$$

Then $\mu_L \geq 0$, $\mathcal{E}_L \geq 0$, and

$$Q_L[f] = \mathcal{E}_L[E_L f] + \gamma_L \|f\|^2 + P_L[f]. \quad (7.32)$$

Proof. For the continuous part of (7.30), (7.31) is (2.4). For the atomic part, write each prime term of (2.9) as $-2c \operatorname{Re}\langle F, U_s F \rangle = -2c\|F\|^2 + 2c[\|F\|^2 - \operatorname{Re}\langle F, U_s F \rangle]$ with $c = (\log p)p^{-m/2}$ and $s = m \log p$; the first terms collect into $\gamma_L - w_0$ and the second into the atomic part of (7.31). Nonnegativity is $\|F\|^2 - \operatorname{Re}\langle F, U_r F \rangle = \frac{1}{2}\|F - U_r F\|^2$. $\square$

Write $S_L[f] = \mathcal{E}_L[E_L f] + \gamma_L \|f\|^2$, so that $Q_L = S_L + P_L$. Away from the diagonal the kernel of S_L is $-\mu_L$, which is nonpositive: S_L is of Stieltjes type, and the single measure μ_L carries the archimedean density and the prime atoms together. This is the presentation in which they are one object rather than two channels, which Theorem 7.5 requires of any source.

Proposition 7.14 (A necessary condition). *Suppose S_L has a normalized ground state ψ with $S_L[\psi] = -\alpha_L$, $\alpha_L > 0$. If $Q_L \geq 0$ on $C_c^\infty(I_L)$ then*

$$\alpha_L \leq 2|\langle \cosh(x/2), f \rangle|^2 \Big|_{f=\psi} \leq 2\|\cosh(x/2)\|_{L^2(I_L)}^2 = 2\sinh(L/2) + L. \quad (7.33)$$

Proof. By (2.17), $0 \leq Q_L[\psi] = -\alpha_L + 2|\langle \cosh(x/2), \psi \rangle|^2 - 2|\langle \sinh(x/2), \psi \rangle|^2$. Drop the last term and apply Cauchy–Schwarz, using $\int_{I_L} \cosh^2(x/2) dx = L/2 + \sinh(L/2)$. $\square$

Remark 7.15 (What the condition is worth, and what it is not). Two things make (7.33) usable. First it is sharp. A scan of 7713 values of L in $[1/2, 10]$ on 4000 cells, with the prime thresholds sampled on both sides, gives least relative margin 9.5×10^{-4} at $L = 5.3396$ and least absolute margin 0.0187; there is no crossing. Second, α_L is bounded from below without diagonalizing anything. The Galerkin value of α_L in any finite basis is such a bound, since a Rayleigh minimum over a subspace is at least the minimum over the space; and in the cell basis $S_L = d_L I - N_L$ with N_L entrywise nonnegative and d_L constant, so α_L is a Perron-root deficit and Collatz–Wielandt gives, for every strictly positive u ,

$$\alpha_L \geq \frac{1}{h} \left(\min_i \frac{(N_L u)_i}{u_i} - d_L \right). \quad (7.34)$$

A positive u making the right side of (7.34) exceed $2\sinh(L/2) + L$ would show that Q_L is not positive, hence that RH fails. The constant function already gives $\alpha_L \geq 3.85$ at $L = 2$, against a threshold of 4.35.

The condition is necessary and not sufficient, and the gap is quantitative rather than technical. It tests Q_L on the one direction in which S_L is negative and the pole form is positive; that direction is close to $\cosh(x/2)$, with cosine 0.995 at $L = 1$ and 0.9998 at $L = 4$. On its orthogonal complement

S_L is bounded below only by its second eigenvalue, which is 2.5×10^{-4} at $L = 2$ and 2.2×10^{-4} at $L = 3$, while the pole form still contributes $-2|\langle \sinh(x/2), f \rangle|^2$, of size up to $2 \sinh(L/2) - L$, which is 0.35 and 1.26 at those two values. The whole difficulty of the criterion lives on that complement, and a bound on an extreme eigenvalue does not reach it.

8 The next complete matching problem

8.1 Relation to spectral realizations of the zeros

Corollary 2.2 places this objective inside a long line of work on spectral interpretations of the zeros: the Hilbert–Pólya expectation as developed by Berry and Keating [13] and surveyed by Sierra [14], the Hamiltonian proposal of Bender, Brody and Müller [15], and the trace formula of Connes on the adèle class space [16, 2]. What distinguishes the present objective is not its conclusion but its class of candidate pairings: protected line and defect algebras whose positive twisted traces are supplied in advance by a field theory, rather than an operator constructed to have the required spectrum. We are not aware of an attempt to realize the Weil form in that class, and the representation-theoretic results used above [5, 6, 7, 8, 9] do not appear to have been applied to it. That is a reason to continue; it is not evidence that the matching exists.

Two features of the existing literature bear directly on Section 7.4. Connes and Consani obtain archimedean Weil positivity from the trace of the scaling action compressed to the orthogonal complement of the range of cutoff projections [17]. That is exactly a mechanism of the kind Proposition 7.3 demands: a negative contact relative to a positive reference arises as $\|(1 - \Pi)v\|^2 = \|v\|^2 - \|\Pi v\|^2$, never as an added channel. Their functional is, however, the archimedean local term with the two pole contributions removed, and their hypothesis $\hat{g}(i/2) = 0$ is by Lemma 2.3 the statement that the source vector lies on a null line of the hyperbolic pole pairing. The mechanism is therefore a precedent for the shape of a subtraction and not a tool for (8.1), which asks for a norm on all of $C_c^\infty(I_L)$ and so cannot discard the pole directions. With Proposition 7.11 the comparison changes shape: that inequality is no longer a prerequisite for anything here, and what would be wanted from this direction is the mechanism rather than the inequality — a compression of a natural positive operator by projections whose rank grows, which is what the spectrum of Π recorded there looks like. Whether it transfers, with the pole directions retained and the primes carried by the compression, is not settled in this manuscript. Bombieri’s analysis of the quadratic functional [19] likewise treats contact, poles and primes together rather than as separate positive pieces. Both are precedents for the structure that Theorem 7.5 shows to be unavoidable.

8.2 What the exclusion leaves

The contact must arise by projection, and now does. A positive norm produces a constant more negative than any of its channels only through a subtraction that is itself a norm, that is through $\|(1 - \Pi)v\|^2$ for a projection Π , or more generally through a compression. Both the gamma and prime data must then live in one space in which the projection acts, rather than in separate positive summands. Theorem 7.8 supplies the arithmetic side of such a presentation: the target is the prime-free form A_L of (7.19) minus a sum of mirror references, and by Corollary 7.9 the Weil criterion on I_L is the single domination $\sum_p \tilde{B}_{r_p, d_p} \preceq A_L$. What remains is one compression, and by Proposition 7.11 its dominating side may be taken to be the gamma energy together with two elementary terms, which is a form one can write down. Everything else — the contact, the odd half of the pole term, and every prime — is then carried by the compression alone.

A prime-free archimedean inequality is the first target. Positivity of (7.19),

$$K[E_L f] + \left(w_0 + \sum_{p < e^L} \tilde{\kappa}_{r_p, d_p} \right) \|f\|^2 + P_L[f] \geq 0, \quad (8.1)$$

is implied by $Q_L \geq 0$, so it cannot fail unless RH does; it is the statement a source for A_L must satisfy, and primes enter it only through the constant. It is *not* needed by Proposition 7.11, whose two sides are positive term by term, so it is no longer a prerequisite for the construction problem. We record its status because it is a natural question about A_L , and because it is not uniform in L .

For $L \leq \log 2$ no prime is active, the constant is w_0 , and (8.1) is $Q_L \geq 0$; there is no slack to be had from the mirror contact. That case is a theorem of Yoshida [18, Theorem 1], whose Hermitian form is (2.9) including both pole terms, who imposes no side condition, and whose test space contains $C_c^\infty(I_L)$; the proof is a computer-assisted numerical argument. The restatement of that theorem in [17] carries the additional hypothesis $\hat{f}(\pm i/2) = 0$, which by Lemma 2.3 is exactly $P_L[f] = 0$, and is the form the result takes once the pole terms are moved out of the functional.

For large L an elementary argument suffices.

Proposition 8.1. *Set $c_L = w_0 + \sum_{p < e^L} \tilde{\kappa}_{r_p, d_p}$ and $N(a) = \int_a^\infty n_\gamma(r) dr = \sum_{n \geq 0} e^{-an} / a_n$. If*

$$c_L + 2N(L/2) \geq 2 \sinh(L/2) - L, \quad (8.2)$$

then (8.1) holds on $C_c^\infty(I_L)$.

Proof. Splitting (2.4) according to whether the second variable lies in I_L , and using $E_L f = 0$ outside, $K[E_L f] = D_L[f] + \int_{I_L} |f|^2 h$ with $D_L \geq 0$ and $h(x) = N(L/2 - x) + N(L/2 + x)$. Since N is convex and decreasing, $h \geq 2N(L/2)$. By (2.17) the pole term is at least $-2|\langle \sinh(x/2), f \rangle|^2$, and $\|\sinh(x/2)\|_{L^2(I_L)}^2 = \sinh(L/2) - L/2$, so Cauchy–Schwarz gives $P_L[f] \geq -(2 \sinh(L/2) - L)\|f\|^2$. Adding the three bounds gives (8.2). $\square$

On each interval between consecutive primes c_L is constant, $N(L/2)$ decreases and $2 \sinh(L/2) - L$ increases, so (8.2) need only be checked at the right endpoints. It first holds on the interval of primes through 11 and holds at every later endpoint below 10^6 , with least margin 0.556 at $L = \log 13$; beyond 10^6 partial summation with $\theta(t) \geq t(1 - 1/\log t)$ gives $\sum_{p < X} \tilde{\kappa}_{r_p, d_p} \geq 1.855\sqrt{X} + 62.9$ against a requirement of $\sqrt{X} + O(\log X)$. Hence (8.1) holds for every $L \geq \log 11$, and a weighted form of the same estimate, keeping the blow-up of h at the endpoints, extends this to $L \geq \log 7$. The remaining range is the compact window $\log 2 < L < \log 7$, where the least Galerkin eigenvalue of A_L is between 0.25 and 1. The supporting computations and the two estimates are recorded in the investigation’s note on this inequality and in `check_prime_free_archimedean.py`.

The pole term needs no separate mechanism. P_L is the contribution of the two poles of ζ , at the non-real points $\tau = \pm i/2$, and every pairing used in this manuscript is positive by construction, so no positive channel adjoined to the others supplies its negative direction. That is Proposition 7.3, and it is a statement about channels. It does not upgrade into a structural requirement on a realization: a realization realizes Q_L , not P_L as a separate summand, and by Corollary 2.2 one compatible in L is unitarily equivalent to the evaluation map on the zeros, in a positive ℓ^2 which has no null vectors at all. The two poles are not vectors of that space. They are the two points the functional equation adds to the zero set, and in (2.13) the term P_L is exactly what makes the right-hand side a sum over zeros only.

What absorbs the pole term is the same domination that absorbs the negative contact. By (2.17) its indefiniteness is an odd-sector, rank-one phenomenon of size at most $2\|\sinh(x/2)\|_{L^2(I_L)}^2 =$

$2\sinh(L/2) - L$, and Proposition 7.11 places it on the source side, inside K_+ . It is not close to binding. In the Galerkin computation of `check_prime_free_archimedean.py` the least eigenvalue of A_L lies in the *even* sector for $L \leq \log 2$, where by (2.17) the pole form is positive; above $\log 2$ the minimum moves to the odd sector, and admitting the negative direction there moves the least eigenvalue from 0.2654 to 0.2567 at $L = 1$ and from 2.31 to 2.12 at $L = 2$.

What Lemma 2.3 and Proposition 2.4 give a candidate is therefore not a sector to look for but a symmetry to check. Every source carries the reflection involution J with $J\Phi(f) = \Phi(Rf)$ and $J^2 = \text{id}$, the parities decouple, and a pairing that does not commute with $x \mapsto -x$ is excluded before any arithmetic is computed. The hyperbolic description of P_L remains the right way to see which combination of the two pole evaluations carries the negative direction; it is not a demand for an indefinite metric.

What to compute in a candidate model. By Corollary 2.2 a realization compatible in L is translation covariant with spectral measure the zero counting measure. The object to extract from a candidate theory is therefore a spectral measure or density of states, and the question to ask of it is whether it is that measure. Reproducing the terms of (2.9) one at a time is not a weaker version of the same question: Theorem 7.5 shows that every term can be matched exactly while no realization exists.

8.3 A finite-prime phase identity is useful background

For comparison with earlier local-factor approaches, define $S_p = J_p/\overline{J_p}$ using (5.5), and set $S_{\mathcal{P}} = S_{\infty} \prod_{p \in \mathcal{P}} S_p$ for a finite prime set. Logarithmic differentiation gives

$$-i\overline{S_{\mathcal{P}}}S'_{\mathcal{P}} = b(\tau^2) + w_0 - 2 \sum_{p \in \mathcal{P}, m \geq 1} (\log p) p^{-m/2} \cos(m\tau \log p). \quad (8.3)$$

When $\mathcal{P}$ contains the active primes of I_L , integration against $|\widehat{E_L f}|^2/(2\pi)$ and addition of $P_L[f]$ gives $Q_L[f]$. Inactive repetitions vanish by support. This argument uses no infinite Euler product on the critical line.

The identity is classical local-factor algebra. Its phase derivative is negative at $\tau = 0$, and the signed poles are outside it. It is neither a positive norm nor an identification of a canonical semilocal positive pairing with the Weil form. The present inverse problem is to find a physical source for the whole form, not merely for this phase.

8.4 A precise open problem

Problem 8.2 (Joint arithmetic preparation). Specify a positive quantum, boundary or defect model, with a physically defined source $\Phi_L(f)$ on smooth compact support, for which the entire pairing is (2.9). Derive the source's adjoint, domain and normalization, including the fixed contact, both signed poles, every active prime repetition, and absence or exact cancellation of mixed translations. For an all-support realization, give one compatible rule as L increases.

A candidate preparation must in addition pass (7.9) for every decomposition into channels whose norms it computes separately, and must produce the pole term and the contact from one mechanism rather than from independent positive pieces. Theorem 7.8 and Proposition 7.11 reduce Problem 8.2 to a single compression, which is worth stating on its own.

Problem 8.3 (The compression). Exhibit the contraction of Proposition 7.11 from a specified model: realize K_+ of (7.23) as the norm of a physically defined source, in place of (7.25), and T_L

of (7.24) as $\langle G \cdot, \Pi G \cdot \rangle$ for a positive contraction Π derived from that model's own structure. By (7.26) this realizes Q_L as a compression, and $\|\Pi\| \leq 1$ is the Weil criterion on I_L , so the contraction produced must be critical rather than strict. The presentation of Corollary 7.9, with A_L on the dominating side, is an alternative form of the same problem; it additionally requires (8.1).

This is an open problem, not an existence conjecture established for a named field theory. A worthwhile intermediate theorem could be conditional on construction of a credible specified theory, provided the construction assumptions do not prescribe the desired covariance. Proposition 3.2 would then give the forward implication to RH. Neither a reverse implication nor an equivalence with Yang–Mills is proposed here.

The two leading directions have complementary roles. Sphere boundary modules give the clearest real Gaussian and oscillator normalization control. The next sphere object should be a boundary or vortex correspondence acting on that control with a derived infinite source. An arbitrary choice of coefficients in an orthogonal basis would not establish the required physical correspondence. The Schur direction now has a specified positive prime source, and its next object should be a charge-neutral gauge or RG/interface preparation.

In the conformal $SU(2)$ benchmark, the magnetic generator has two shifted contributions and a neutral coefficient, schematically

$$H = uC_+(v)u + C_0(v) + u^{-1}C_-(v)u^{-1}, \quad (8.4)$$

and the representation includes a Weyl constraint [6, Sections 5.1.2–5.1.3]. The notation in (8.4) describes the structure to retain; it is not yet a specified arithmetic source. The exact coefficients and projection must be carried through a chosen neutral line or interface word. The neutral term and projection are possible mechanisms for joint interference, not terms whose desired effect can be assumed. This calculation must also implement the real-input lift or replace it by another derived boundary map.

The immediate target is the complete identity at $L = 1$:

$$\|\Phi_1(f)\|^2 = K[E_1f] + w_0\|f\|^2 + P_1[f] - \frac{2\log 2}{\sqrt{2}} \operatorname{Re}\langle E_1f, U_{\log 2}E_1f \rangle. \quad (8.5)$$

The gamma response, contact, pole terms and prime atom must be computed together, and by Theorem 7.5 they cannot be computed apart. In the form of Proposition 7.11 the same test reads: realize $K[E_1f] + 2|\langle \cosh(x/2), f \rangle|^2$ as a physically defined source norm, and $(-w_0 - \tilde{\kappa}_{r_2, d_2})\|f\|^2 + 2|\langle \sinh(x/2), f \rangle|^2 + \tilde{B}_{r_2, d_2}[E_1f]$ as a compression of it. At $L = 5/4$ the atom at $\log(3/2)$ must be absent; at $L = 3/2$ the second return of 2 must have primitive coefficient $(\log 2)/2$ for each oriented translation, while the prime 3 term remains active. These are complete identity tests. Failure calls for a different source or model rather than a growing sequence of residual estimates.

8.5 Scope of the working manuscript

The positive gamma action, its integer refinement, and the positive prime reference are established ingredients. The dressed calculation of Section 6 places the last of these in an explicit elementary Schur preparation. Section 7.4 then shows that this preparation cannot be completed in the way (7.7) suggests: no coherent source has the gamma energy and the summed prime references as its two channel norms. Section 7.5 replaces that architecture by a subtraction in which the source is prime-free and every prime translation sits in a compression. The source tests in this manuscript are sufficiently explicit to be checked and reused, but they have not supplied the full matching identity. The main remaining difficulty has therefore not been reduced to a numerical sign check.

Other constructions remain available: interacting boundary gluing, integrable defects, gauge networks and arithmetic graph models may provide different source rules. Their broader literature comparison is retained in the investigation's research notes, while this working manuscript focuses on the calculations above. The evidence for sphere and Schur systems is the availability of explicit protected representations and positive pairings, not an assertion that their arithmetic matching is inevitable. Further drafts should strengthen a specific joint construction or replace it when its complete pairing fails.

Preparation

Substantial parts of this working manuscript were produced with a large language model, Anthropic's Claude (model `claude-opus-5`), working directly in the investigation repository: the literature comparison, the check programs of Appendix D, and first drafts of Sections 2.5, 7.4, 7.5, 6.5 and 7.6, together with the revision of Section 8.2 that withdraws the indefinite-metric requirement stated in the previous version. In version 0.6 the same model corrected the saturation values of Section 7.6, which the previous version attributed to a computation that does not produce them, added Remark 7.12, and drafted Section 7.7. The research notes accompanying the investigation name the model used for each. Every statement here is classified in the text as a written proof, a one-sided numerical certificate, or a labelled floating-point illustration, and Appendix D states what the programs do and do not establish.

A A positive gamma action and its integer refinement

This appendix records the common positive bulk from which a different joint source could be built. For $\alpha > 0$, let

$$a_n(\alpha) = 2(n + \alpha), \quad b_\alpha(s) = \sum_{n \geq 0} \frac{2}{a_n(\alpha)} \frac{s}{a_n(\alpha)^2 + s}. \quad (\text{A.1})$$

Thus $b = b_{1/4}$. For line fields u_n with weak derivatives, define the nonnegative joint energy

$$\mathcal{E}_\alpha(F, \{u_n\}) = \sum_{n \geq 0} \frac{2}{a_n(\alpha)} \left(\|F - u_n\|^2 + a_n(\alpha)^{-2} \|u'_n\|^2 \right). \quad (\text{A.2})$$

The finite-energy domain is specified by the sum as a whole. In particular, the divergent expression $\sum_n 2\|F\|^2/a_n(\alpha)$ must not be split off as a separate energy from compensating fields.

Proposition A.1 (Elimination and positive refinement). *For smooth compactly supported F ,*

$$\inf_{\{u_n\}} \mathcal{E}_\alpha(F, \{u_n\}) = \frac{1}{2\pi} \int_{\mathbb{R}} b_\alpha(\tau^2) |\widehat{F}(\tau)|^2 d\tau. \quad (\text{A.3})$$

For each integer $p \geq 2$, put $\alpha_j = (\alpha + j)/p$, $0 \leq j < p$, and $D_p F(x) = p^{-1/2} F(x/p)$. Then

$$b_\alpha(s) = \frac{1}{p} \sum_{j=0}^{p-1} b_{\alpha_j}(s/p^2), \quad (\text{A.4})$$

$$\mathcal{E}_\alpha(F, \{u_n\}) = \frac{1}{p} \sum_{j=0}^{p-1} \mathcal{E}_{\alpha_j}(D_p F, \{D_p u_{pm+j}\}_{m \geq 0}). \quad (\text{A.5})$$

The latter identity is valid as an equality of nonnegative extended energies, before eliminating the fields.

Proof. For each mode, minimizing the Fourier-space quadratic function gives $\hat{u}_n = a_n^2(a_n^2 + \tau^2)^{-1}\hat{F}$ and the summand in (A.1). Substitution of all minimizers, or finite towers followed by monotone convergence, proves (A.3). For refinement, write $n = pm + j$. Then $a_{pm+j}(\alpha) = p a_m(\alpha_j)$, $\|D_p u\| = \|u\|$, and $\|(D_p u)'\| = p^{-1}\|u'\|$. Substitution proves (A.5); reindexing the positive multiplier sum gives (A.4). $\square$

The first examples are

$$\begin{aligned} b_{1/4}(s) &= \tfrac{1}{2}b_{1/8}(s/4) + \tfrac{1}{2}b_{5/8}(s/4), \\ b_{1/4}(s) &= \tfrac{1}{3} \left[b_{1/12}(s/9) + b_{5/12}(s/9) + b_{3/4}(s/9) \right]. \end{aligned} \quad (\text{A.6})$$

Refining by p and then by another integer ℓ gives residue labels $j + pk$, the dilation $D_{p\ell}$ and weights $1/(p\ell)$. Thus refinement by 2 then 3, or 3 then 2, gives the same six branches $\{1, 5, 9, 13, 17, 21\}/24$ up to permutation.

There is also a contact identity. Set $m_\alpha(\tau) = \text{Re } \psi(\alpha + i\tau/2) - \log \pi$. Gauss multiplication [3] gives

$$m_\alpha(\tau) = \log p + \frac{1}{p} \sum_{j=0}^{p-1} m_{\alpha_j}(\tau/p). \quad (\text{A.7})$$

Subtracting the value at zero recovers (A.4). Only the kinetic actions in (A.2) have the asserted positivity; (A.7) is not a positive decomposition of the complete archimedean form.

These formulas exist for every integer, and so do not select primes. Coordinate dilation by p is also different from translation by $\log p$. A conservative junction acting on the common refined fields could be tested, but its injection, return and readout maps would have to derive these relations. The normalization $p^{-1/2}$ in the residual-output inclusion is fixed by (A.5); it is not yet a physical arithmetic return amplitude. Refinement or a unitary rotation of all outputs preserves their norm and cannot itself introduce prime terms. No novelty claim is made for gamma multiplication or its semilocal use [4].

B Gauge transfer and two first-prime source tests

B.1 Winding a holonomy is different from transfer evolution

Two-dimensional heat-kernel Yang–Mills supplies a rigorous control with explicit Wilson functionals [11]. Normalize the positive $SU(2)$ Casimir so that the spin- j representation has value $j(j+1)$. For dimensionless area-coupling parameter $s > 0$,

$$\mathbb{E}_s[\chi_j(U)] = (2j+1)e^{-sj(j+1)}. \quad (\text{B.1})$$

The normalized fundamental loop wound m times is $w_m(U) = \frac{1}{2} \text{Tr}_{\mathbb{C}^2}(U^m) = \cos(m\vartheta)$, for eigenvalues $e^{\pm i\vartheta}$. The character identity $2 \cos(m\vartheta) = \chi_{m/2} - \chi_{(m-2)/2}$ for $m \geq 2$ gives

$$\mathbb{E}_s[w_1] = e^{-3s/4}, \quad \mathbb{E}_s[w_m] = \frac{1}{2} \left[(m+1)e^{-sm(m+2)/4} - (m-1)e^{-sm(m-2)/4} \right]. \quad (\text{B.2})$$

If the first return is set to $r \in (0, 1)$, the second is $(3r^{8/3} - 1)/2$, strictly less than r^2 . Indeed, for $y = r^{2/3}$,

$$3y^4 - 2y^3 - 1 = (y-1)(3y^3 + y^2 + y + 1) < 0. \quad (\text{B.3})$$

No overall primitive-length coefficient repairs this relative failure.

The same literal winding rule fails for every central probability convolution semigroup with proposed moments $\mathbb{E}_t[w_1] = r_t = e^{-\lambda t}$ and $\mathbb{E}_t[w_2] = r_t^2$, $\lambda > 0$. The normalized spin-one character coefficient would be $c_1(t) = (1 + 2r_t^2)/3$. Convolution requires multiplicativity, but

$$c_1(t+u) - c_1(t)c_1(u) = \frac{2}{9}(1 - r_t^2)(1 - r_u^2) > 0 \quad (t, u > 0). \quad (\text{B.4})$$

In contrast, let $H = \frac{2}{3}C_2$ on the character Hilbert space with Haar inner product, and choose the unit vector $\eta = \chi_{1/2}$. Then $H\eta = \frac{1}{2}\eta$ and

$$\langle \eta, e^{-m(\log p)H} \eta \rangle = p^{-m/2}. \quad (\text{B.5})$$

This propagates a fixed representation for a longer interval; it does not wind the same holonomy. It passes the scalar repetition test.

Even so, the literal affine Euclidean source $\Phi_L(f) = \int_{I_L} f(x) e^{-(b+x)H} \eta \, dx$, $b > L/2$, has a Hankel kernel $C(2b+x+y)$, where $C(t) = \langle \eta, e^{-tH} \eta \rangle$. If $H \geq 0$ and this kernel is invariant under small simultaneous translations of x, y , then C' vanishes on an interval. Its spectral measure satisfies $-C'(t) = \int \lambda e^{-t\lambda} \, d\nu_\eta(\lambda)$, so η must lie in the zero-energy subspace. The resulting kernel is constant. This excludes that affine-time source, not other spatial, normalized or extended gauge preparations.

B.2 A positive disk system for masses and returns

Let Λ be the Dirichlet-to-Neumann operator on the unit disk. On the reflection-even boundary subspace, with normalized measure $d\vartheta/(2\pi)$, use the basis $e_0 = 1$, $e_n = \sqrt{2} \cos(n\vartheta)$ for $n \geq 1$. Harmonic extension gives $\Lambda e_n = n e_n$. Thus

$$A = 2\Lambda + \frac{1}{2}\text{id}, \quad A e_n = a_n e_n, \quad (\text{B.6})$$

has the gamma spectrum and positive quadratic form

$$\langle h, Ah \rangle = \frac{2}{2\pi} \int_{\mathbb{D}} |\nabla u_h|^2 \, d^2 z + \frac{1/2}{2\pi} \int_{S^1} |h|^2 \, d\vartheta. \quad (\text{B.7})$$

Here u_h is the harmonic extension; the form domain is the even $H^{1/2}$ trace space. The choice of half-unit shift and one mode per nonnegative integer is explicit model data, not forced by an arithmetic symmetry.

The transfer is a shifted Poisson semigroup. On the full circle it also has the positive subordination formula

$$e^{-tA} = e^{-t/2} \frac{t}{\sqrt{\pi}} \int_0^\infty s^{-3/2} e^{-t^2/s} e^{-sD^2} \, ds, \quad D = -i\partial_\vartheta, \quad (\text{B.8})$$

which restricts to the even subspace. The scalar Laplace integral proves the formula on each Fourier mode. This is a change of transfer generator, not an equivalence with the original Yang–Mills dynamics.

Let $P_0 = |e_0\rangle\langle e_0|$ and $T_p = e^{-(\log p)A}$. Then

$$b(\tau^2) = 2 \, \text{Tr} \left[A^{-1} \frac{\tau^2}{A^2 + \tau^2} \right],$$

$$\text{Tr}(P_0 T_p^m) = p^{-m/2}, \quad T_p T_\ell = T_{p\ell}, \quad \text{Tr}(T_p^m) = \frac{p^{-m/2}}{1 - p^{-2m}}. \quad (\text{B.9})$$

The ground projection is therefore essential to the proposed pure geometric return. This model does not derive primality, the primitive coefficient $\log p$, or the absence of mixed histories.

B.3 An orthogonal first-prime branch has an unavoidable cusp

Use the source (4.6) and set $a = 1/2$, $d = \log 2$, $r = e^{-ad} = 2^{-1/2}$. With $H_x = -\partial_x^2$, consider the full positive preparation

$$F \mapsto \left(\{\mathcal{A}_n F\}_{n \geq 1}, \sqrt{1-\theta} \mathcal{A}_0 F, \sqrt{\theta} \mathcal{A}_0 (\text{id} - r U_d) F \right), \quad 0 \leq \theta \leq 1. \quad (\text{B.10})$$

Its Gram operator is, in form notation, $K + \theta(2/a)H_x(a^2 + H_x)^{-1}[r^2 \text{id} - r(U_d + U_d^*)]$. Matching the prime atom forces $\theta = ad/2 = d/4$. The resulting complete pairing has operator

$$B = K + dr^2 \text{id} - dr(U_d + U_d^*) + R_{a,d}, \quad R_{a,d} = -da^2[r^2 \text{id} - r(U_d + U_d^*)](a^2 + H_x)^{-1}. \quad (\text{B.11})$$

The resolvent kernel $e^{-a|s|}/(2a)$ gives

$$R_{a,d}(s) = \frac{ad}{2} \begin{cases} r^2 e^{a|s|}, & |s| \leq d, \\ e^{-a|s|}, & |s| \geq d. \end{cases} \quad R'_{a,d}(d+) - R'_{a,d}(d-) = -da^2 r \neq 0. \quad (\text{B.12})$$

On $d < L < \log 3$, after the common prime atom is removed, the candidate-minus-target difference is $(dr^2 - w_0) \text{id} + E_L^* R_{a,d} E_L - P_L$. The pole kernel is smooth near $x - y = d$ and the contact has no support there. Neither can cancel the cusp, so the full pairing cannot match on that interval. This is an exact kernel obstruction, requiring no residual bound.

For a finite positive sum of analogous orthogonal branches with mass a_n , positive weight θ_n and attenuation r_n , the derivative jump at d is $-2\theta_n a_n r_n$ for each active branch. The signs agree and cannot cancel. General coherent changes of source are outside this assertion.

B.4 A coherent repair cancels the cusp and still fails the contact

In two common outputs, use the lowest gamma source and a shift source

$$\Gamma_0 F = \begin{pmatrix} 2H_x(a^2 + H_x)^{-1} F \\ \partial_x(a^2 + H_x)^{-1} F \end{pmatrix}, \quad Z_{u,v} F = \begin{pmatrix} u(U_d - U_d^*) F \\ vF + u(U_d + U_d^*) F \end{pmatrix}, \quad a = \frac{1}{2}, \quad (\text{B.13})$$

where u, v are real. All operators commute on the line. The first components are respectively self-adjoint and skew-adjoint, and the second components respectively skew-adjoint and self-adjoint. Hence

$$\Gamma_0^* Z + Z^* \Gamma_0 = 0, \quad Z^* Z = (v^2 + 4u^2) \text{id} + 2uv(U_d + U_d^*). \quad (\text{B.14})$$

Keeping the other gamma modes and setting $2uv = -dr$ gives

$$B_{\text{coh}} = K + C \text{id} - dr(U_d + U_d^*), \quad C = v^2 + 4u^2 \geq 2dr. \quad (\text{B.15})$$

The inequality follows from $C - 2dr = (v + 2u)^2$. There is no displaced continuous term, but on the first-prime interval

$$B_{\text{coh},L} - Q_L = (C - w_0) \text{id} - P_L \neq 0. \quad (\text{B.16})$$

Indeed $C - w_0 > 0$ while P_L has rank two on an infinite-dimensional input space. This coherent representation is still the sum of independent gamma and jump Gram forms, because its Hermitian interference vanishes. It supplies a second exact stopping criterion for the tested source.

C A finite rational-feedback obstruction

A common attempt to fit a negative contact is to filter a positive reference source by finitely many rational functions of its Gram operator. The following elementary argument excludes a specific class without estimating its residual.

Proposition C.1. *Let A_0 be a closed source into a positive Hilbert space and let $T = A_0^* A_0 > 0$ have infinitely many distinct eigenvalues. On finite linear combinations of eigenvectors, form the direct-sum source*

$$A_{\text{rat}} f = (A_0 r_1(T) f, \dots, A_0 r_m(T) f, s_1(T) f, \dots, s_k(T) f), \quad (\text{C.1})$$

where all r_j, s_ℓ are rational functions with complex coefficients, finite at every eigenvalue. For any $\alpha > 0$, this source cannot satisfy

$$\|A_{\text{rat}} f\|^2 = \langle f, (T - \alpha \text{id}) f \rangle \quad (\text{C.2})$$

on all eigenvectors, even if $T - \alpha \text{id}$ is positive on its spectrum. Positive constant metrics within either group of channels can be absorbed into the rational functions and obey the same conclusion.

Proof. For an eigenvalue λ , the proposed identity gives $\lambda \sum_j |r_j(\lambda)|^2 + \sum_\ell |s_\ell(\lambda)|^2 = \lambda - \alpha$. Write $r^\#(z) = \overline{r(\bar{z})}$ and define the rational function

$$R(z) = z \sum_j r_j(z) r_j^\#(z) + \sum_\ell s_\ell(z) s_\ell^\#(z). \quad (\text{C.3})$$

Clearing denominators, infinitely many distinct roots force $R(z) = z - \alpha$ as a rational identity. Choose a real x with $0 < x < \alpha$ outside the finite pole set. Its sum-of-squares representation gives $R(x) \geq 0$, while $x - \alpha < 0$, a contradiction. $\square$

No finite accumulation point of eigenvalues is needed. A nonzero polynomial has only finitely many roots. Finite cascades of scalar rational filters of the same reference remain in this class. For example,

$$\lambda \left(1 - \frac{\kappa}{\lambda + \mu} \right)^2 = \lambda - 2\kappa + \frac{2\kappa\mu}{\lambda + \mu} + \frac{\kappa^2\lambda}{(\lambda + \mu)^2}, \quad (\text{C.4})$$

whose last terms are nonnegative for $\lambda, \kappa > 0, \mu \geq 0$. Adding finitely many channels of (C.1) cannot remove the exact constant in this reference-only class.

The proposition does not exclude general finite-field models, nonrational responses from continuum dynamics, couplings to operators other than functions of T , infinite channel families, cross couplings outside (C.1), or agreement on only finitely many inputs. It is a scoped selection result with no assertion of literature priority.

D Checks, provenance and mathematical status

The analytical arguments in this draft are written proofs and explicitly qualified construction problems. Ten small standard-library Python programs accompany the investigation. They check finite algebra and normalization identities rather than positivity of the Weil form. The preserved records distinguish exact rational checks from labelled floating-point illustrations.

Program in numerics/	Cases	Scope
<code>check_matching.py</code>	153	Rational filters and gamma refinement
<code>check_gauge_transfer.py</code>	326	Winding, transfer and first-prime algebra
<code>check_sphere_schur.py</code>	251	Charged norms, product ratios and undressed overlaps
<code>check_dressed_schur.py</code>	587	Dressed norm, magnetic coefficients and mixed-return witnesses
<code>check_explicit_formula.py</code>	8	Normalization against the sum over zeta zeros
<code>check_channel_bound.py</code>	75	Interference bound and the two-channel certificates
<code>check_mirror_subtraction.py</code>	75	Mirror weight, subtraction identity, domination quotients
<code>check_prime_free_archimedean.py</code>	78,666	Parity split, pole factorisation, the criterion of Proposition 8.1 at every prime breakpoint below 10^6
<code>check_mirror_dressing.py</code>	638	Reachable output states, the mirror dressing, and its weight
<code>check_gamma_compression.py</code>	32,037	The identity $Q_L = K_+ - T_L$, the sign change of c_L , and one-sided saturation

The 112,816 finite cases are supporting checks, not 112,816 independent theorems. Finite-product end factors and truncation boundary coefficients are retained. The finite mixed-return sample does not replace the unique-factorization proof in Proposition 7.2. The Mellin and beta integrals, infinite-series convergence, graph-domain argument and kernel exclusions are justified in the text; the programs do not formally verify them.

The explicit-formula, channel-bound, mirror-subtraction and prime-free programs are labelled floating-point computations rather than exact rational ones. `check_explicit_formula.py` evaluates both sides of (2.13) for the input $F(x) = \frac{1}{2}(1 + \cos(2\pi x/L))$ on I_L , whose transform is $\hat{F}(\tau) = \omega^2 \sin(\tau L/2)/(\tau(\omega^2 - \tau^2))$ with $\omega = 2\pi/L$. It computes the zero ordinates itself, from the Hardy function through Borwein's series for η , and agrees with published values to 10^{-13} . For $L = 2, 3, 4$ the two sides agree to 2.5×10^{-10} , 5.4×10^{-11} and 1.0×10^{-11} , against totals of 2.9×10^{-5} , 2.6×10^{-6} and 1.4×10^{-7} : the ratio of the largest single term of (2.9) to the total rises from 1.4×10^5 to 6.3×10^7 across those three intervals. This check confirms the conventions of (2.3) as a whole. It is not evidence for RH, since its ordinates are computed on the critical line by construction. `check_channel_bound.py` evaluates (7.11) for step-function inputs, for which every quantity is a finite closed form; the one infinite series, $\sum_n(\cdot)e^{-a_n r}$, is summed after subtracting its exactly known leading part, and the reported remainder bound for the two certificates of Theorem 7.5 is below 10^{-40} . `check_mirror_subtraction.py` verifies (7.15) and (7.16) in exact rational arithmetic, checks (7.18) as an identity of quadratic forms on step inputs, and reports the domination quotients discussed after Corollary 7.9. Those quotients are one sided: a value above one on a subspace certifies that a domination fails, a value below one certifies nothing. The Galerkin eigenvalues quoted in Section 7.5 are labelled floating-point computations and are not part of the replayed record.

`check_prime_free_archimedean.py` builds the forms of (2.9) and (7.19) in a uniform cell basis, checks that basis against the independent step routine of the channel-bound program, verifies (2.17) and (2.16) on random step inputs, and evaluates (8.2) at all 78,497 prime breakpoints below 10^6 . Its Galerkin quotients are one sided in the same sense as the mirror-subtraction ones. Two of its outputs are conclusive rather than indicative, because a negative value in a subspace certifies that a form is not positive: the indicator of I_L gives $K[E_L 1] + w_0 \|1\|^2 < 0$ for $L \leq 3/2$, so the pole term of

(8.1) cannot be dispensed with; and an explicit step function on $I_{3/4}$ gives $K + w_0\|\cdot\|^2 + P_L$ the value $-3.2 \times 10^{-3}\|f\|^2$, so the archimedean form with the bare constant w_0 and no mirror contact is not positive for any $L \geq 3/4$. Since $\tilde{\kappa}_{r_2, d_2} = 0.574$ switches on at $L = \log 2$, the mirror contact of Theorem 7.8 is what keeps (7.19) positive beyond that point.

`check_mirror_dressing.py` works with truncated power series in exact rational arithmetic. It verifies (6.17) for both the dressing of Lemma 6.1 and the mirror dressing (7.21), the ratio of the two electric states, the telescoping of the vacuum Pochhammer at four truncations, the Laurent coefficients of the two weights in the ratios $2/(1+r)$ and $2/(1-r)$ with geometric ratio r , the orthogonality of the two output states — the overlap truncated at N is exactly r^{2N} — and the reachability of three targets unrelated to the prime references. Its one floating-point item, labelled as such, is the geometric rate r at which the truncated dressings' images converge at the radius q^{-1} where the magnetic action evaluates them.

`check_gamma_compression.py` verifies Proposition 7.11 as a matrix identity on indicator cells at ten values of L between $1/2$ and 5 , agreeing to 7.5×10^{-16} , and rebuilds the mirror matrices independently, checking them against (7.18). It locates the sign change of c_L at $p = 13$ exactly, confirms $T_L \geq 0$ at every sampled L , and reports the saturation of the domination. That last column is one sided in the direction opposite to a Rayleigh minimum: a subspace value is a *lower* bound for $\|\Pi\|$, so a value meaningfully above one would certify that the domination fails and a value below one certifies nothing. The values quoted in Section 7.6 come from a Fourier–Galerkin space, which resolves the near-degeneracy better than the cell basis recorded here.

The manuscript is self-contained at the level of its stated external inputs: the classical Weil criterion and the cited model representation data. It does not input a shared background file or cite equation numbers from one. The accompanying coverage map relates its sections to the retained research notes, including the broader candidate survey. Build instructions and a hash record identify the reviewed sources and compiled PDF. The `drafts/` folder preserves complete dated source-and-PDF snapshots. Those hashes establish file identity, not mathematical correctness or portability of PDF metadata between TeX installations.

The following boundaries of the results are essential. The positive prime reference is known; the dressed theorem gives an explicit elementary Schur source for it. Square-summable formal magnetic outputs do not settle all closed-operator extensions. The continuous lift has not been derived from a four-dimensional interface. The general protected-sector construction hypotheses have not been proved here. Theorem 7.5 excludes one specified architecture and not a class of theories. Most importantly, the full contact-and-pole identity in Problem 8.2 remains unsolved, so this draft makes no claim to establish RH or a completed reduction to constructive QFT.

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